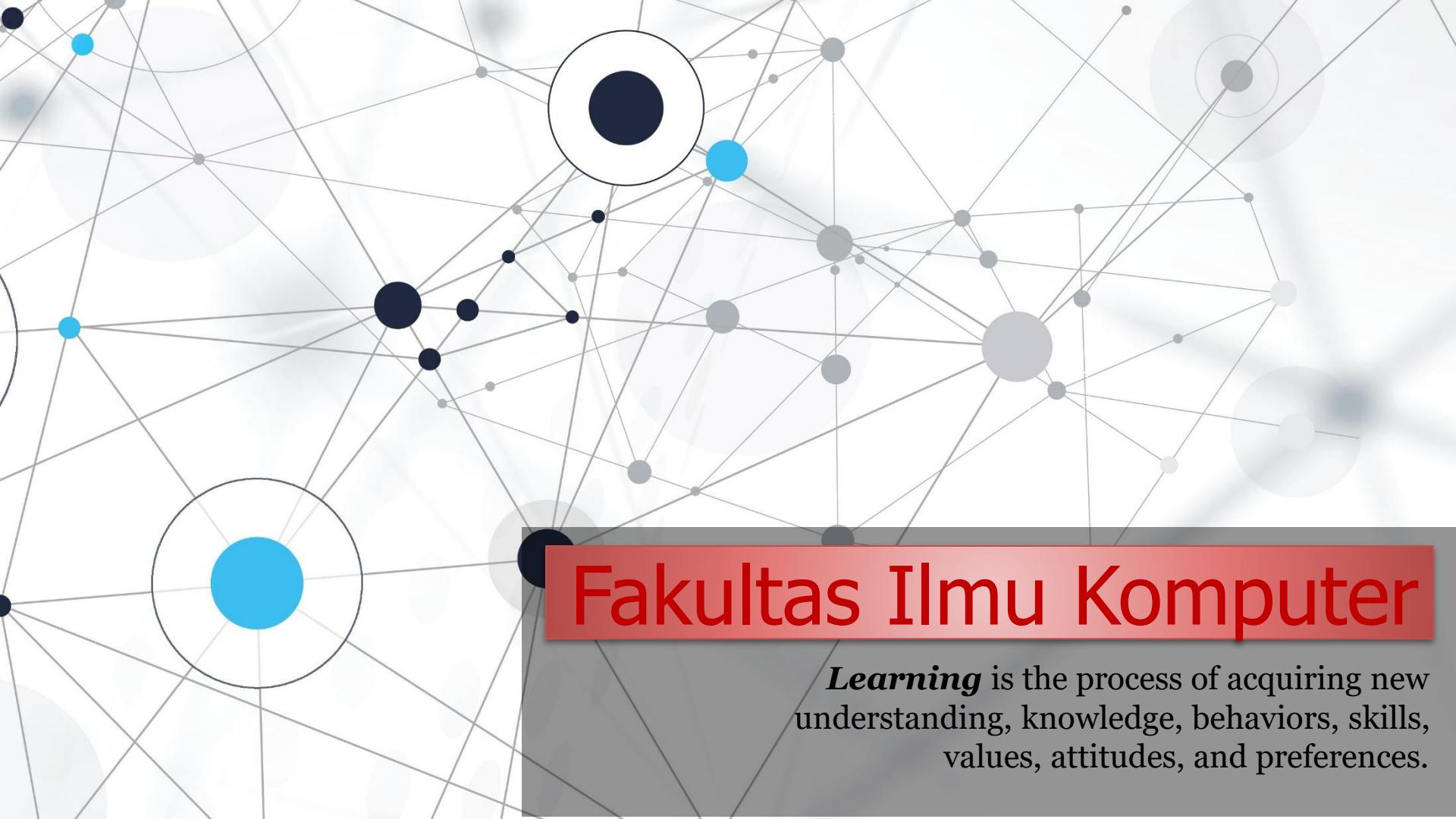




Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Back-end Framework #5

Component View

Amsal 11:17 TB

“Orang yang murah hati berbuat baik kepada diri sendiri, tetapi orang yang kejam menyiksa badannya sendiri”.



**Good or bad,
what you put out
comes back to you!**

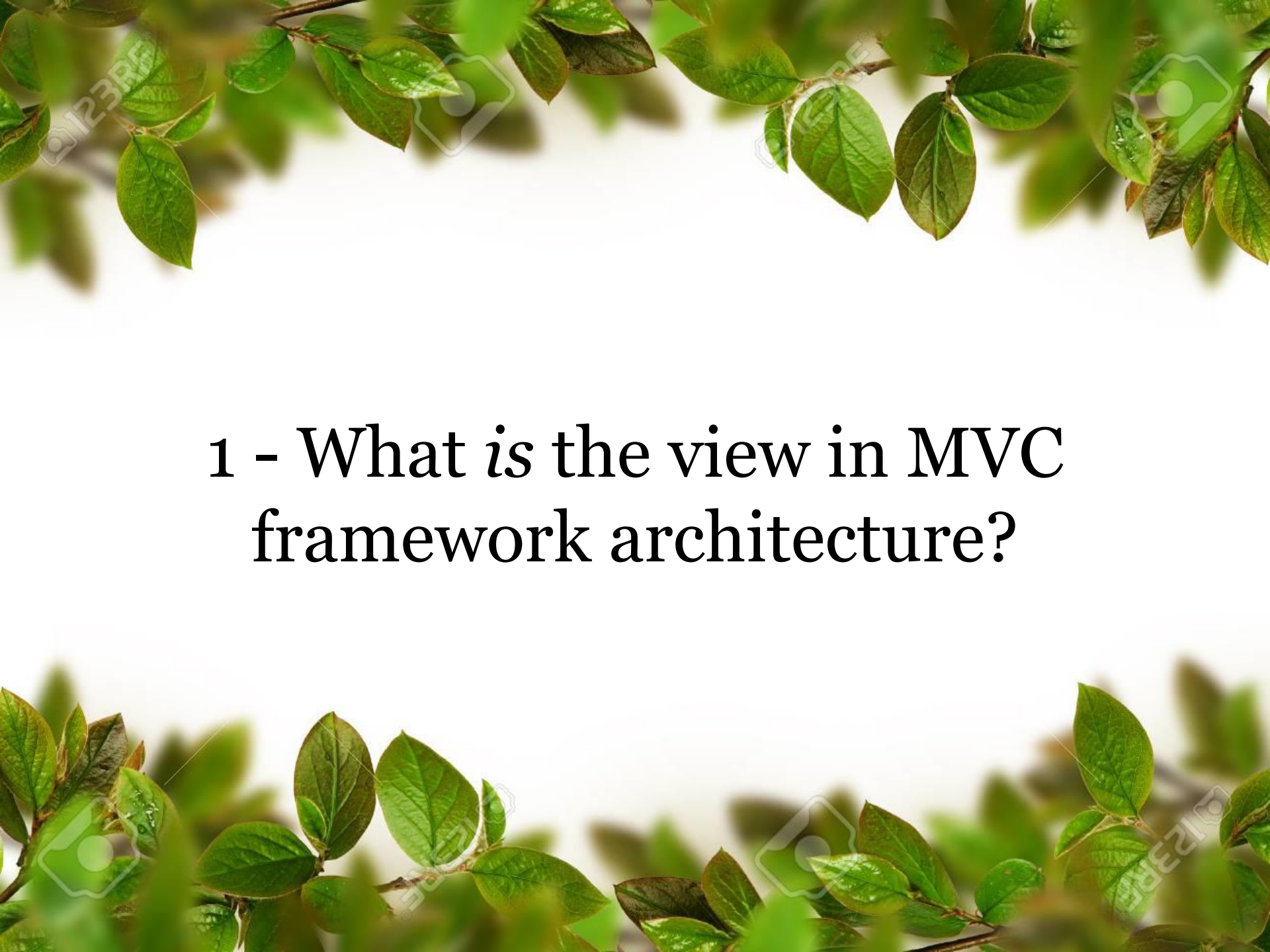
Mengapa harus berbuat baik?

karena pada saat kita berbuat baik, maka kita berbuat baik kepada diri kita sendiri.

1. Berbuat baik adalah suatu keuntungan yang tak terelakkan. Dengan kata lain, saat kita *berbuat baik kita merasa baik.*
2. Hal ini bukan berarti kita merasa Hebat, puas diri atau membenarkan diri.
3. Namun hal ini berarti kita mengalami kelembutan dan ketenangan dan merendahkan diri dalam kesadaran bahwa kita menjadi penyalur kasih Allah.
4. Memberi pertolongan kepada orang lain, yaitu dengan melakukan sebatas kemampuan kita.



Welcome *to* the Back-end Web Development *course*

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in the middle of the slide.

1 - What is the view in MVC framework architecture?

Three Important MVC Components (*layers*)

Component #1 → Model :

- ✓ The model component stores data and its related logic. It represents data that is being transferred between controller components or any other related business logic. For example, a Controller object will retrieve the customer info from the database. It manipulates data and send back to the database or use it to render the same data.
- ✓ It responds to the request from the views and also responds to instructions from the controller to update itself. It is also the lowest level of the pattern which is responsible for maintaining data.

Component #2 → View :

- ✓ The View is that part of the application that represents the presentation of data.
- ✓ Views are created by the data collected from the model data. A view requests the model to give information so that it resents the output presentation to the user.
- ✓ The view also represents the data from chats, diagrams, and table. For example, any customer view will include all the UI components like text boxes, drop downs, etc.

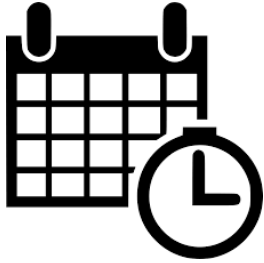
Component #3 → Controller :

- ✓ The Controller is that part of the application that handles the user interaction. The controller interprets the mouse and keyboard inputs from the user, informing model and the view to change as appropriate.
- ✓ A Controller send's commands to the model to update its state (*E.g., Saving a specific document*). The controller also sends commands to its associated view to change the view's presentation (*For example scrolling a particular document*).

The paper was published under the title:

Burbeck, Steve. "Applications programming in smalltalk-80 (*tm*): How to use model-view-controller (mvc)." *Smalltalk-80 v2 5* (1992): 1-11.

Three Important MVC Components (*layers*)



Model: It includes all the data and its related logic.



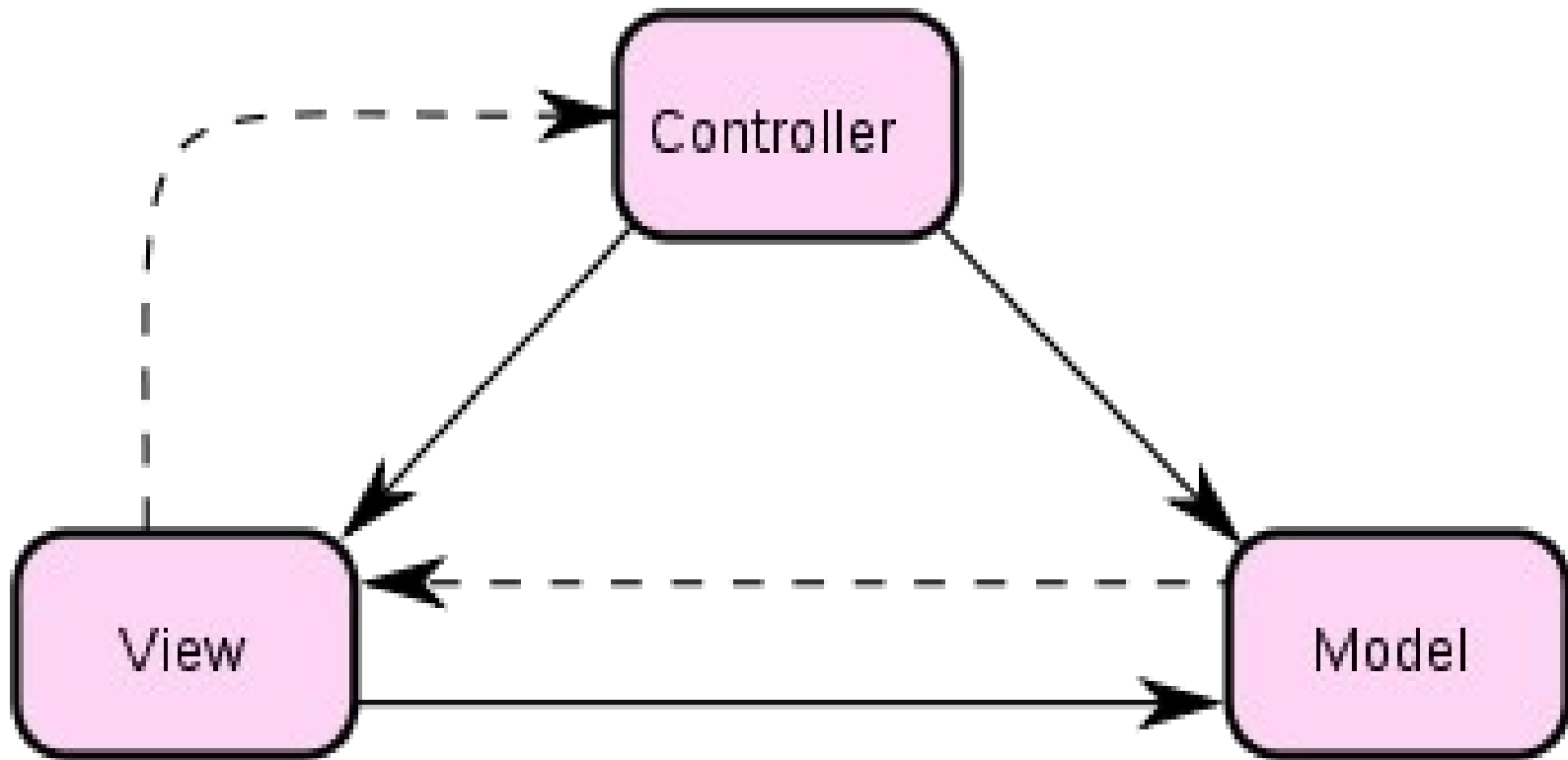
View: Present data to the user or handles user interaction.



Controller: An interface between Model and View components.

Three Important MVC Components (*layers*)

Logic program (*controller*)



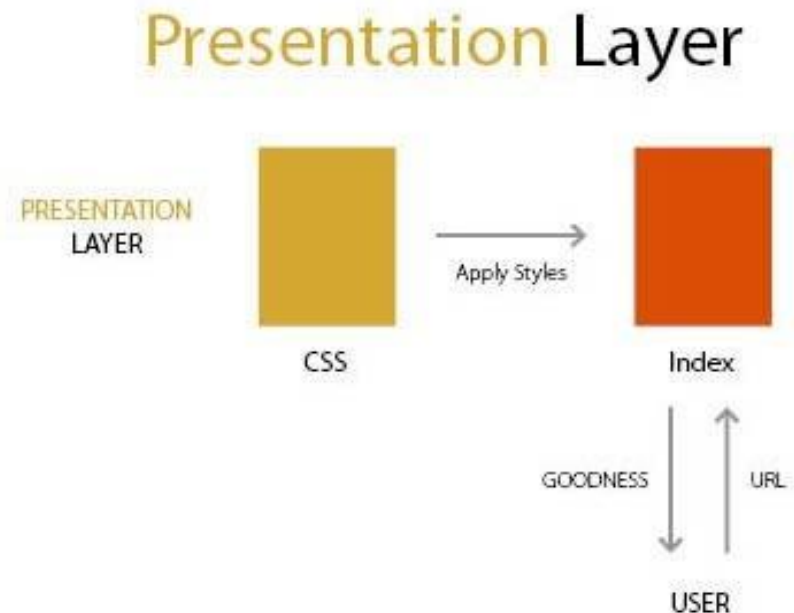
Tampilan (*view*) => UI

Manage the data

Component #2 $\Rightarrow$ View (Presentation)



Responsible to display the data provided by the model in a specific format.



A view is responsible for UI (*user interface*). View displays the data coming from the model.

Component #2 $\Rightarrow$ View (Presentation)



View in MVC is a user interface. A view is used to display data to the user using the model class object and also enables users to modify them.

Types of Views

1) Normal view

It's the same as web page. To create this view we use the view page template, (*MVC to view page template in MVC application*).

2) Partial view

It's the same as user control. Partial views are similar to the user controller.

3) Layout View

It's the same as a master page. To create a master page view we use a template.

Read Source:

<https://tekslate.com/many-types-view-mvc>



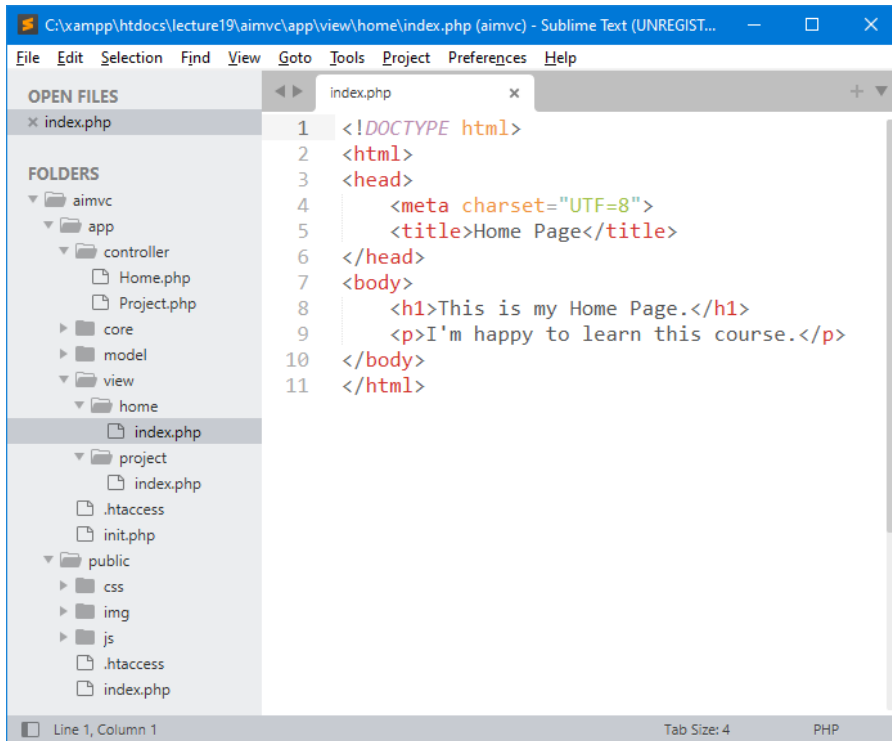
Start *from* Here !!!

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in the middle of the slide.

2 – Basic *display* the page from
“*home & project*” view.

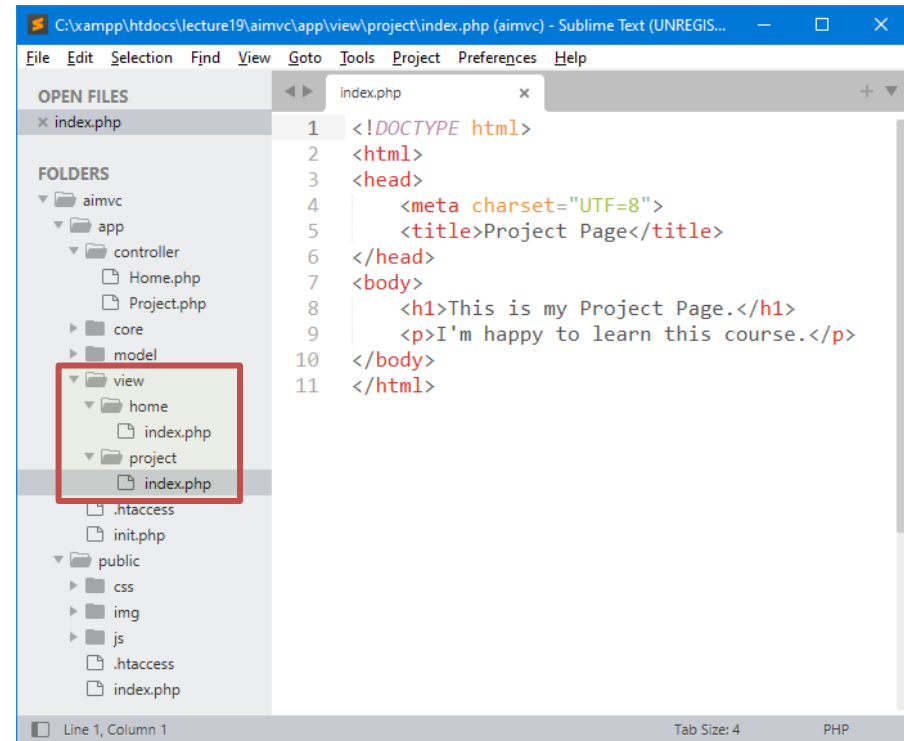
1 - Create Subdirectory in View

/view/home/index.php



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <meta charset="UTF-8">
5   <title>Home Page</title>
6 </head>
7 <body>
8   <h1>This is my Home Page.</h1>
9   <p>I'm happy to learn this course.</p>
10 </body>
11 </html>
```

/view/project/index.php



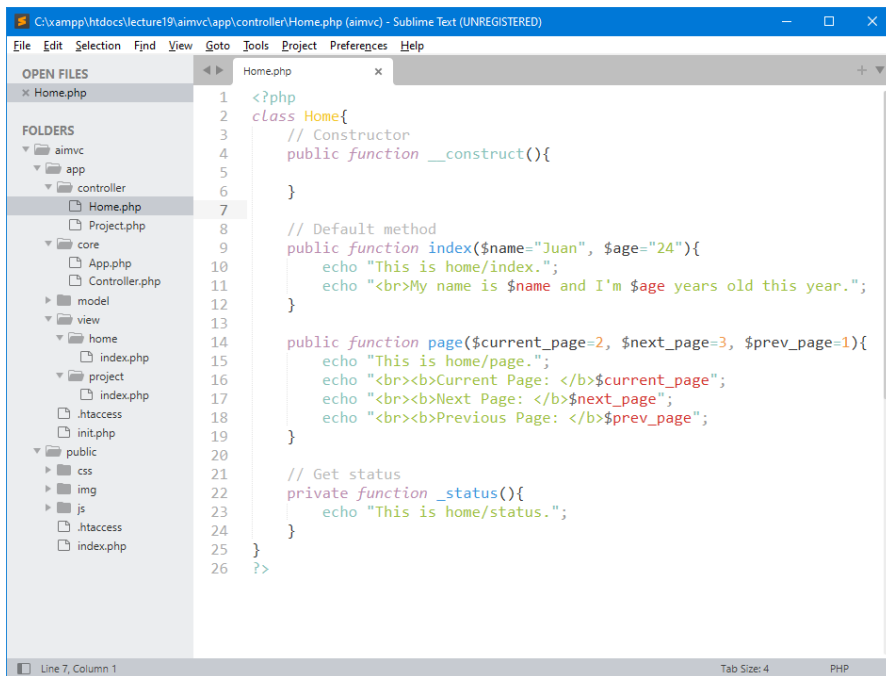
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <meta charset="UTF-8">
5   <title>Project Page</title>
6 </head>
7 <body>
8   <h1>This is my Project Page.</h1>
9   <p>I'm happy to learn this course.</p>
10 </body>
11 </html>
```

- 1) The **<!DOCTYPE html>** declaration defines that this document is an HTML5 document
- 2) The **<html>** element is the root element of an HTML page
- 3) The **<head>** element contains meta information about the HTML page
- 4) The **<title>** element specifies a title for the HTML page (*which is shown in the browser's title bar or in the page's tab*)
- 5) The **<body>** element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- 6) The **<h1>** element defines a large heading
- 7) The **<p>** element defines a paragraph

2 – Display Page from View

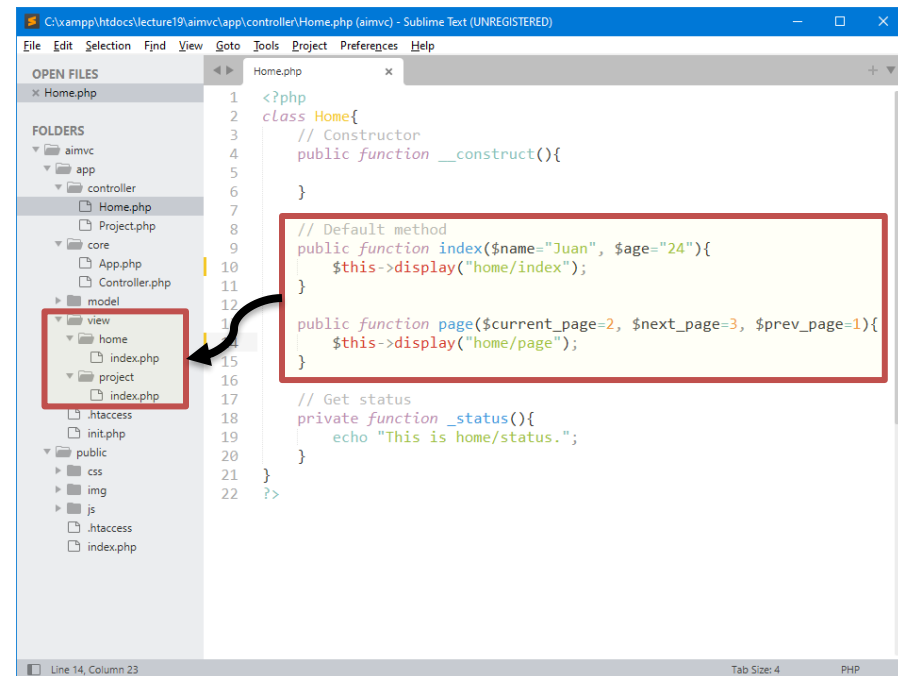
We handle this action from Controller. See, before and after changing code.

Code before change



```
1 <?php
2 class Home{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name="Juan", $age="24"){
10         echo "This is home/index.";
11         echo "<br>My name is $name and I'm $age years old this year.";
12     }
13
14     public function page($current_page=2, $next_page=3, $prev_page=1){
15         echo "This is home/page.";
16         echo "<br><b>Current Page: </b>$current_page";
17         echo "<br><b>Next Page: </b>$next_page";
18         echo "<br><b>Previous Page: </b>$prev_page";
19     }
20
21     // Get status
22     private function _status(){
23         echo "This is home/status.";
24     }
25 }
26 ?>
```

Code after change



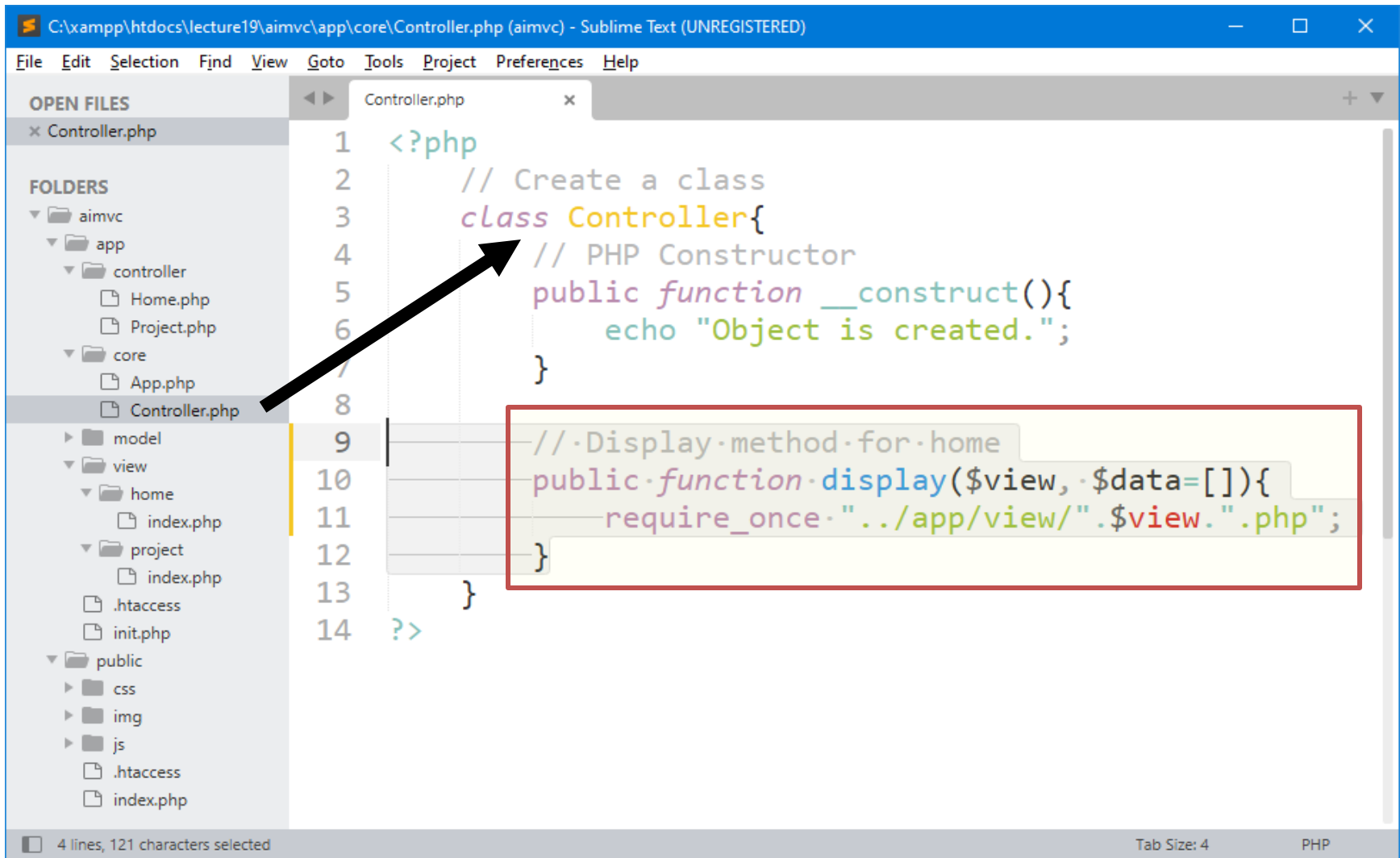
```
1 <?php
2 class Home{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name="Juan", $age="24"){
10         $this->display("home/index");
11     }
12
13     public function page($current_page=2, $next_page=3, $prev_page=1){
14         $this->display("home/page");
15     }
16
17     // Get status
18     private function _status(){
19         echo "This is home/status.";
20     }
21 }
22 ?>
```

\$this->view("home/index");

Note, please see the selection file in left side tree menu. This code means you want to access ***index.php*** in home subdirectory which is located in view directory.

3 – Create Method *display()* in Core Controller

In variable “*view*” may contains value like “*home/index*”. This function will require file from ***view directory*** to get specific page and display it, like ***home*** page.



```
C:\xampp\htdocs\lecture19\aimvc\app\core\Controller.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
Controller.php
x Controller.php
FOLDERS
aimvc
  app
    controller
      Home.php
      Project.php
    core
      App.php
      Controller.php
    model
    view
      home
        index.php
      project
        index.php
      .htaccess
      init.php
    public
      css
      img
      js
      .htaccess
      index.php
1 <?php
2 // Create a class
3 class Controller{
4 // PHP Constructor
5 public function __construct(){
6     echo "Object is created.";
7 }
8
9 // Display method for home
10 public function display($view, $data=[]){
11     require_once "../app/view/" . $view . ".php";
12 }
13 }
14 ?>
```

4 lines, 121 characters selected

Tab Size: 4

PHP

4 – Inherit *display()* Function

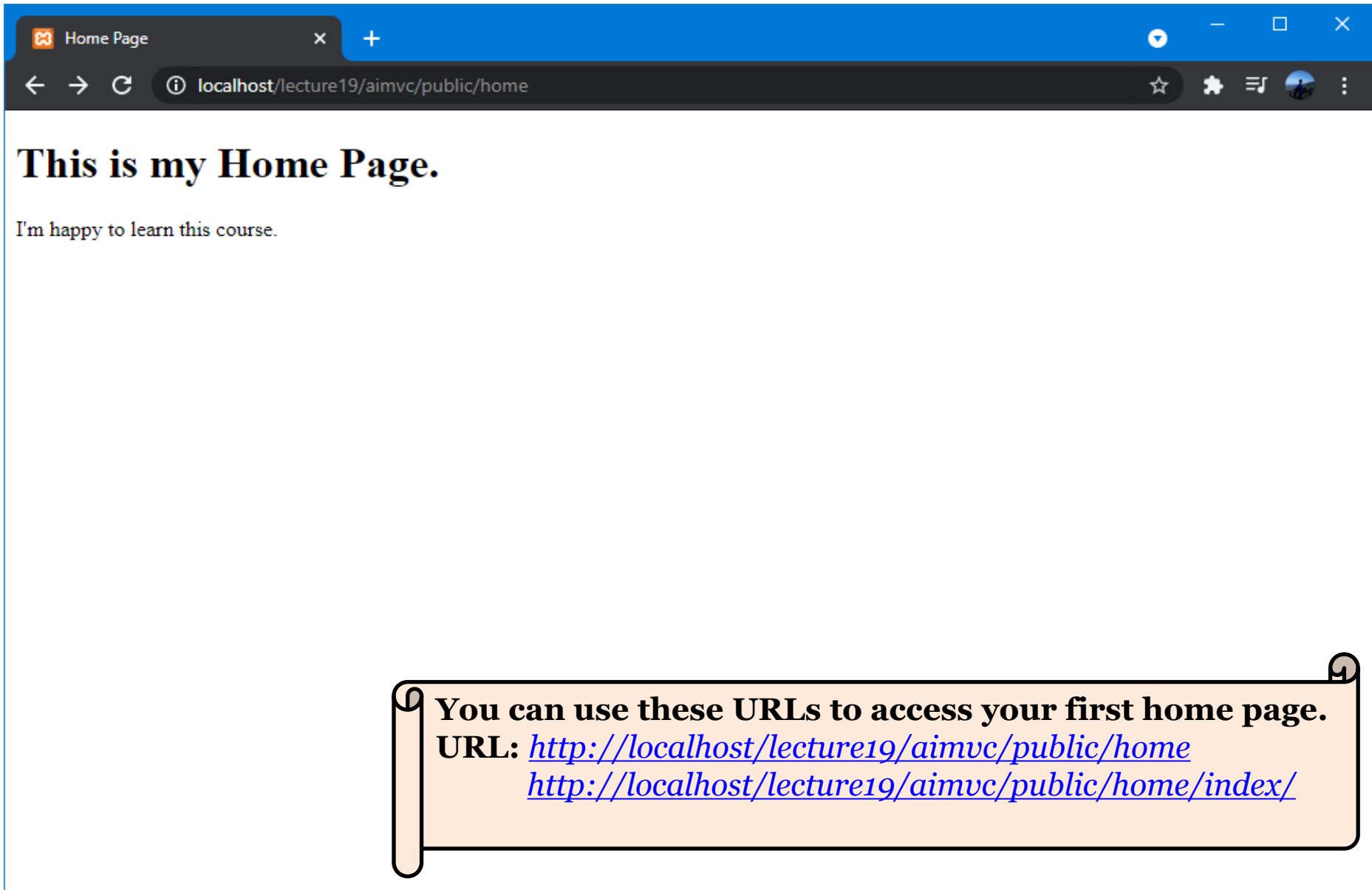
Last step in this stage is to inherit this ***display()* function** to the controller class. The ***extends keyword*** is used to derive a class from another class. This is called inheritance.

```
C:\xampp\htdocs\lecture19\aimvc\app\controller\Home.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
OPEN FILES
x Home.php
FOLDERS
  aimvc
    app
      controller
        Home.php
        Project.php
      core
        App.php
        Controller.php
      model
      view
        home
          index.php
        project
          index.php
      .htaccess
      init.php
      public
        css
        img
        js
        .htaccess
        index.php
  Home.php
1 <?php
2 class Home extends Controller{
3 // Constructor
4 public function __construct(){
6 }
7
8 // Default method
9 public function index($name="Juan", $age="24"){
10 $this->display("home/index");
11 }
12
13 public function page($current_page=2, $next_page=3, $prev_page=1){
14 $this->display("home/page");
15 }
16
17 // Get status
18 private function _status(){
19 echo "This is home/status.";
20 }
21 }
22 ?>
```

30 characters selected Tab Size: 4 PHP

This controller class in core directory inherits all function to home class in controller directory.

5 - Testing – View Home Page



```
<?php
class Home extends Controller{
    // Constructor
    public function __construct(){

    }

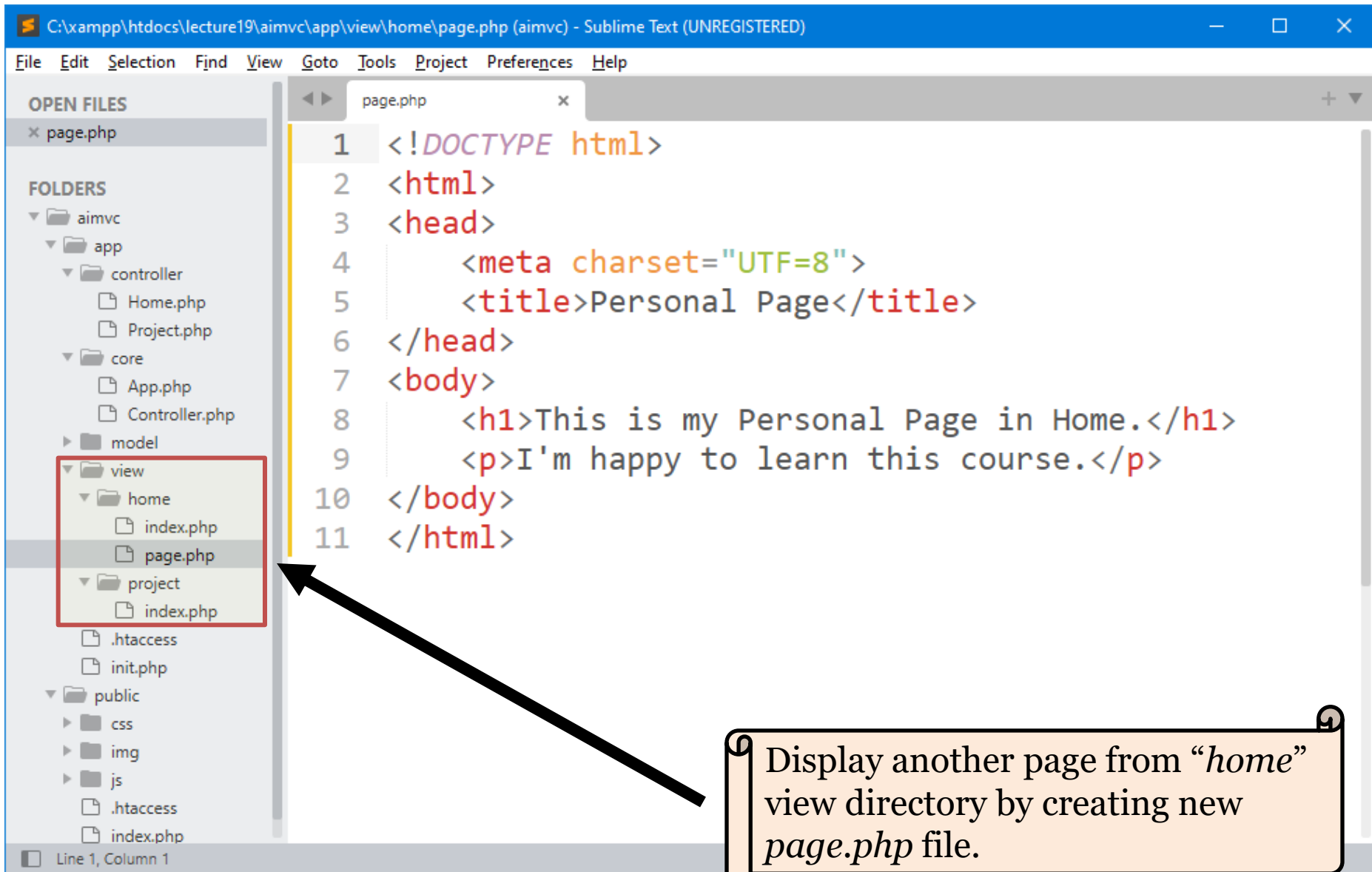
    // Default method
    public function index($name="Juan", $age="24"){
        $this->display("home/index");
    }

    public function page($current_page=2, $next_page=3, $prev_page=1){
        $this->display("home/page");
    }

    // Get status
    private function _status(){
        echo "This is home/status.";
    }
}
```

How to access this method *page()* in the home class? It will display an error if you force to access this method from the URL.

6 – Create *page.php* in *view/home* Directory



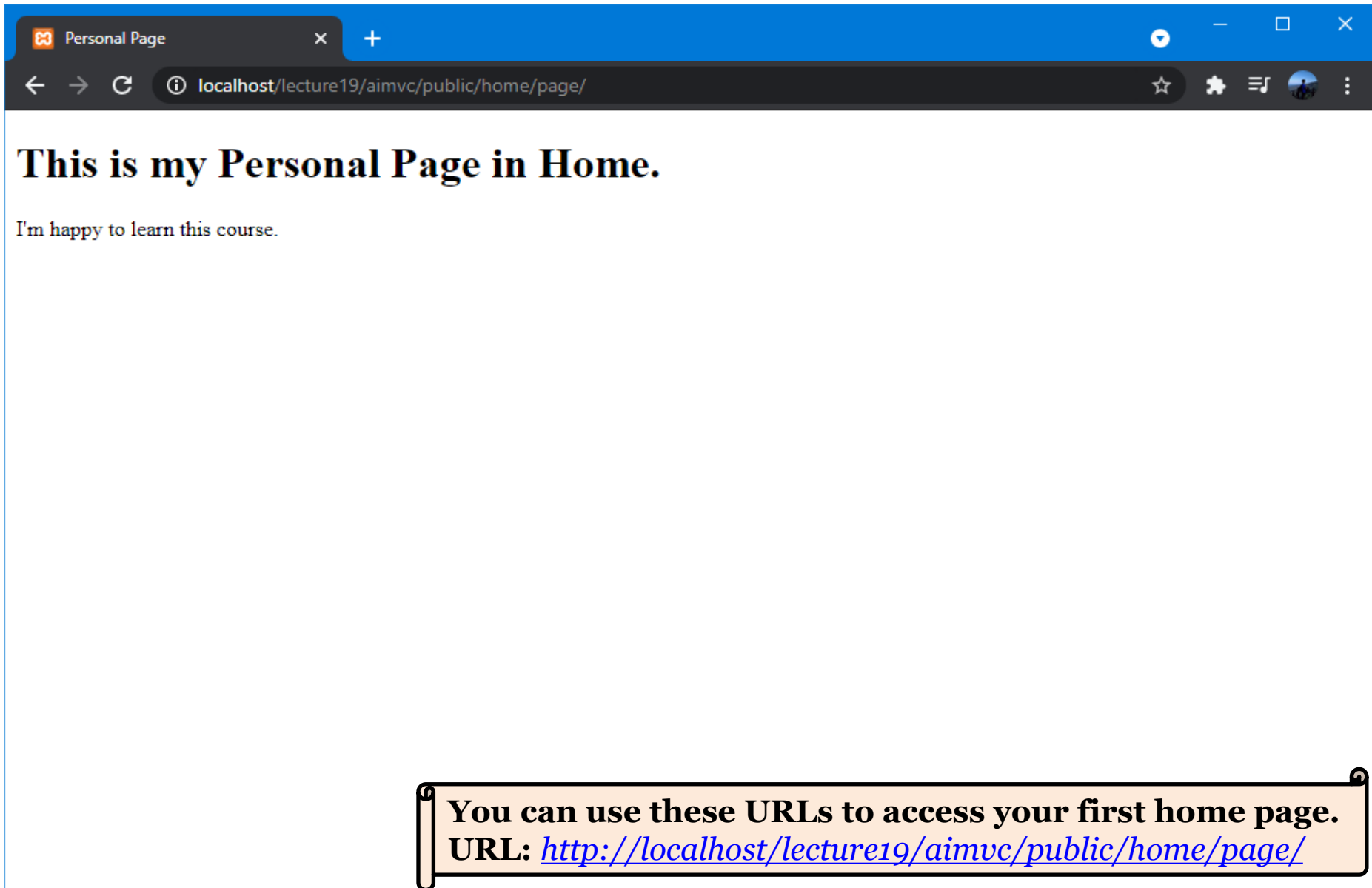
The screenshot shows the Sublime Text editor interface. The title bar indicates the file path: `C:\xampp\htdocs\lecture19\aimvc\app\view\home\page.php (aimvc) - Sublime Text (UNREGISTERED)`. The menu bar includes File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, and Help. The left sidebar displays the 'FOLDERS' tree, where the 'view' directory is expanded, and 'home' is selected, showing 'index.php' and 'page.php'. A red box highlights the 'view' directory, and a black arrow points from a callout box to the 'page.php' file. The main editor area shows the content of 'page.php' with line numbers 1 through 11. The code is as follows:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Personal Page</title>
6 </head>
7 <body>
8     <h1>This is my Personal Page in Home.</h1>
9     <p>I'm happy to learn this course.</p>
10 </body>
11 </html>
```

Line 1, Column 1

Display another page from “home” view directory by creating new *page.php* file.

5 - Testing ⇒ View Personal Page

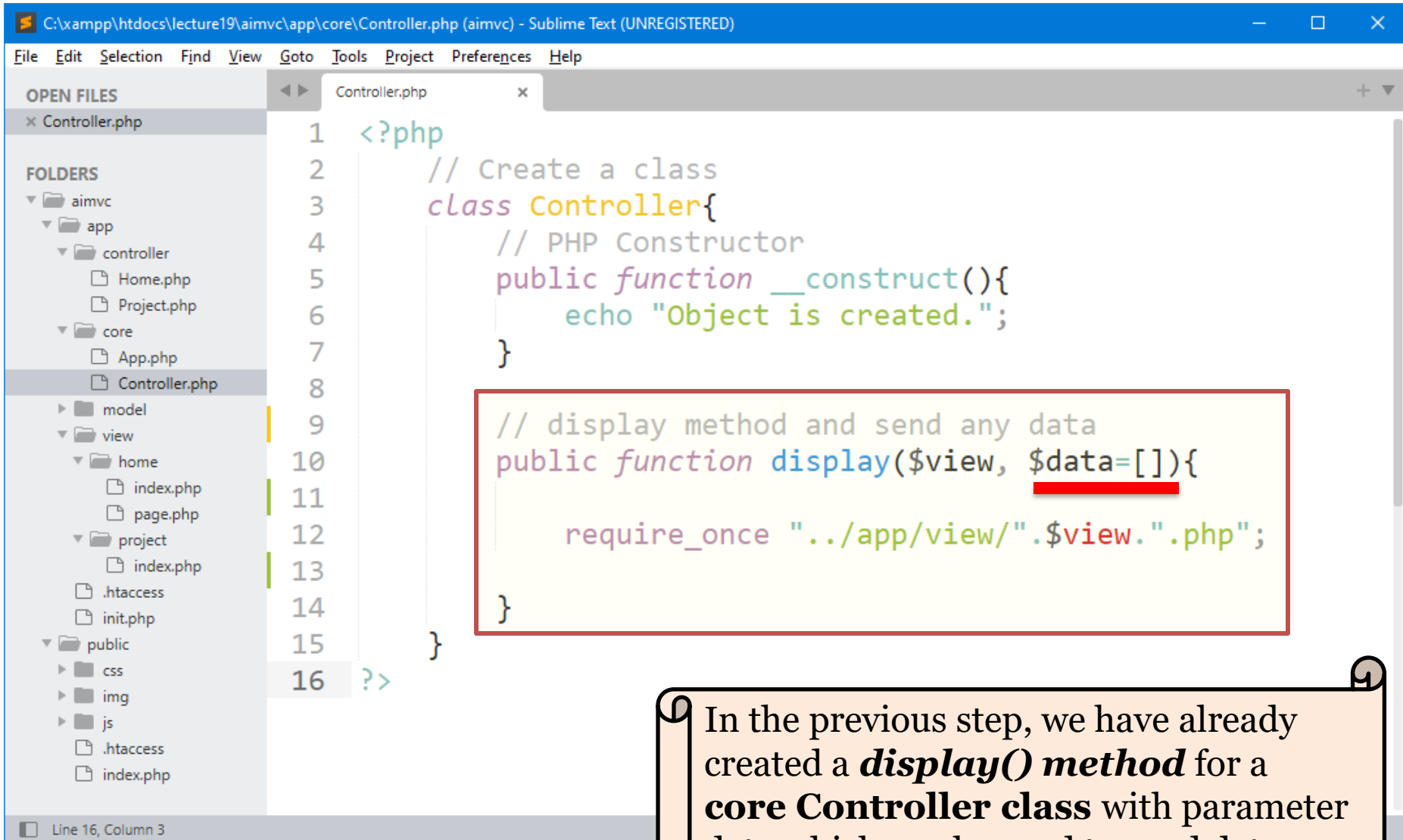


A decorative border of green leaves and branches at the top of the slide.

3 – Send and Receive Data from URL

A decorative border of green leaves and branches at the bottom of the slide.

1 – Check *display()* Method with Parameter Data



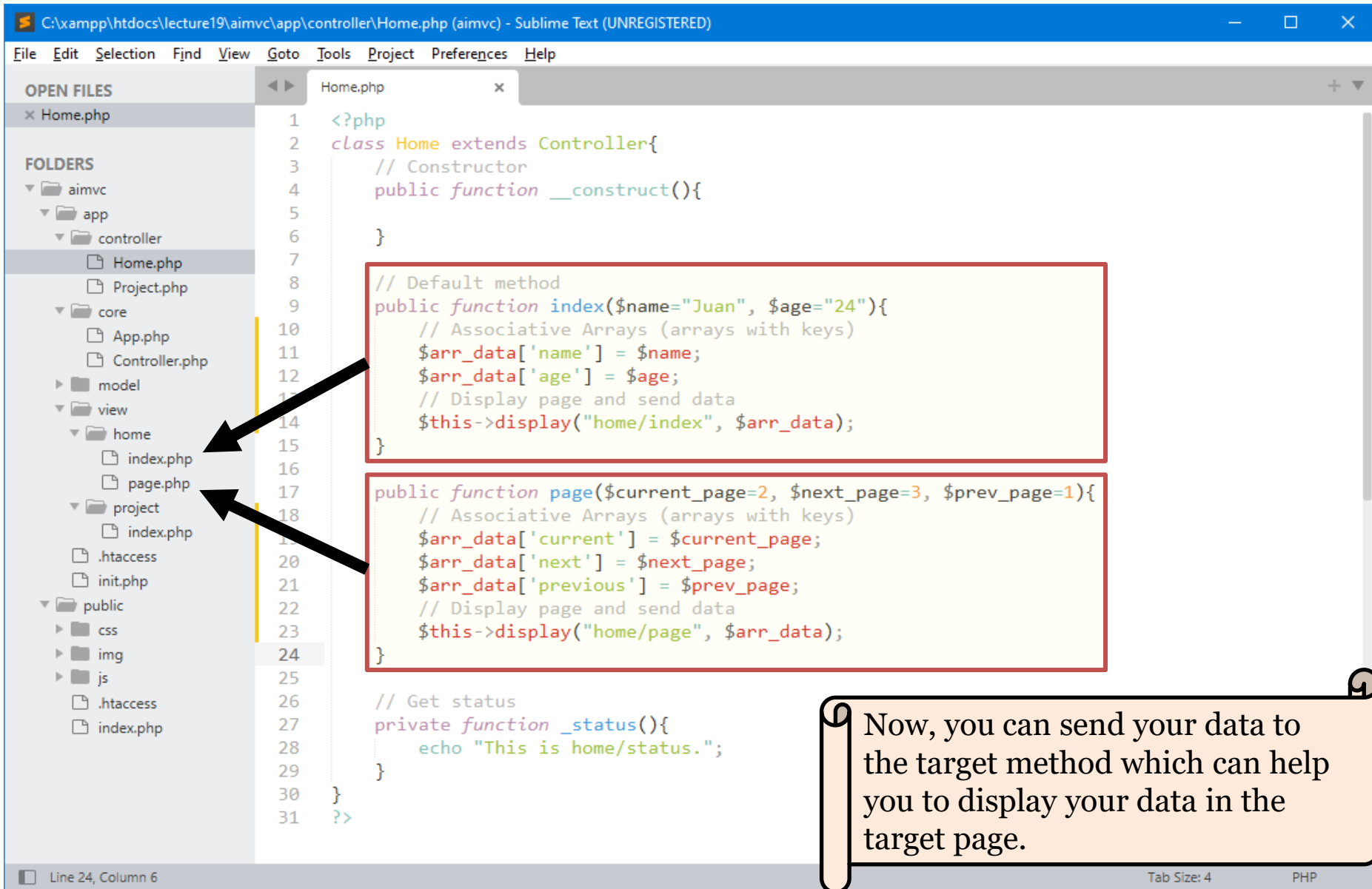
The screenshot shows a Sublime Text editor window titled "C:\xampp\htdocs\lecture19\aimvc\app\core\Controller.php (aimvc) - Sublime Text (UNREGISTERED)". The editor displays the following PHP code:

```
1 <?php
2 // Create a class
3 class Controller{
4     // PHP Constructor
5     public function __construct(){
6         echo "Object is created.";
7     }
8
9     // display method and send any data
10    public function display($view, $data=[]){
11
12        require_once "../app/view/".$view.".php";
13
14    }
15 }
16 ?>
```

The `display` method is highlighted with a red box. A callout box explains its purpose:

In the previous step, we have already created a ***display()* method** for a **core Controller class** with parameter data which can be used to send data.

2 - Assign an Associative Arrays to Method



C:\xampp\htdocs\lecture19\aimvc\app\controller\Home.php (aimvc) - Sublime Text (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES

- × Home.php

FOLDERS

- aimvc
 - app
 - controller
 - Home.php
 - Project.php
 - core
 - App.php
 - Controller.php
 - model
 - view
 - home
 - index.php
 - page.php
 - project
 - index.php
 - public
 - css
 - img
 - js
 - .htaccess
 - index.php

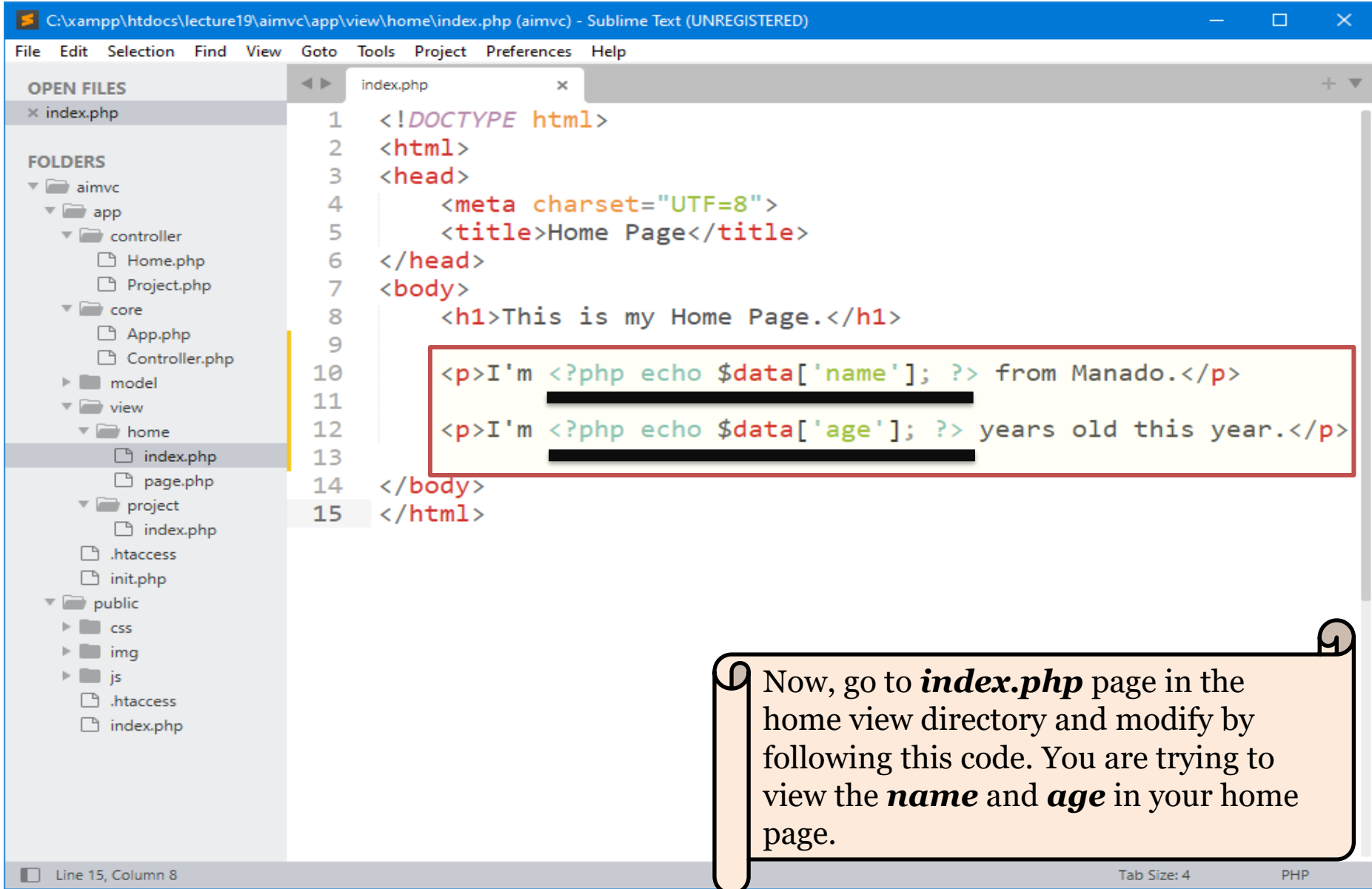
```
1 <?php
2 class Home extends Controller{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name="Juan", $age="24"){
10         // Associative Arrays (arrays with keys)
11         $arr_data['name'] = $name;
12         $arr_data['age'] = $age;
13         // Display page and send data
14         $this->display("home/index", $arr_data);
15     }
16
17     public function page($current_page=2, $next_page=3, $prev_page=1){
18         // Associative Arrays (arrays with keys)
19         $arr_data['current'] = $current_page;
20         $arr_data['next'] = $next_page;
21         $arr_data['previous'] = $prev_page;
22         // Display page and send data
23         $this->display("home/page", $arr_data);
24     }
25
26     // Get status
27     private function _status(){
28         echo "This is home/status.";
29     }
30 }
31 ?>
```

Now, you can send your data to the target method which can help you to display your data in the target page.

Line 24, Column 6

Tab Size: 4 PHP

3 – Receive data from *display()* Parameter



The screenshot shows a Sublime Text editor window titled "C:\xampp\htdocs\lecture19\aimvc\app\view\home\index.php (aimvc) - Sublime Text (UNREGISTERED)". The editor displays the contents of the `index.php` file, which is an HTML document with embedded PHP code. The code is as follows:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Home Page</title>
6 </head>
7 <body>
8     <h1>This is my Home Page.</h1>
9
10    <p>I'm <?php echo $data['name']; ?> from Manado.</p>
11
12    <p>I'm <?php echo $data['age']; ?> years old this year.</p>
13
14 </body>
15 </html>
```

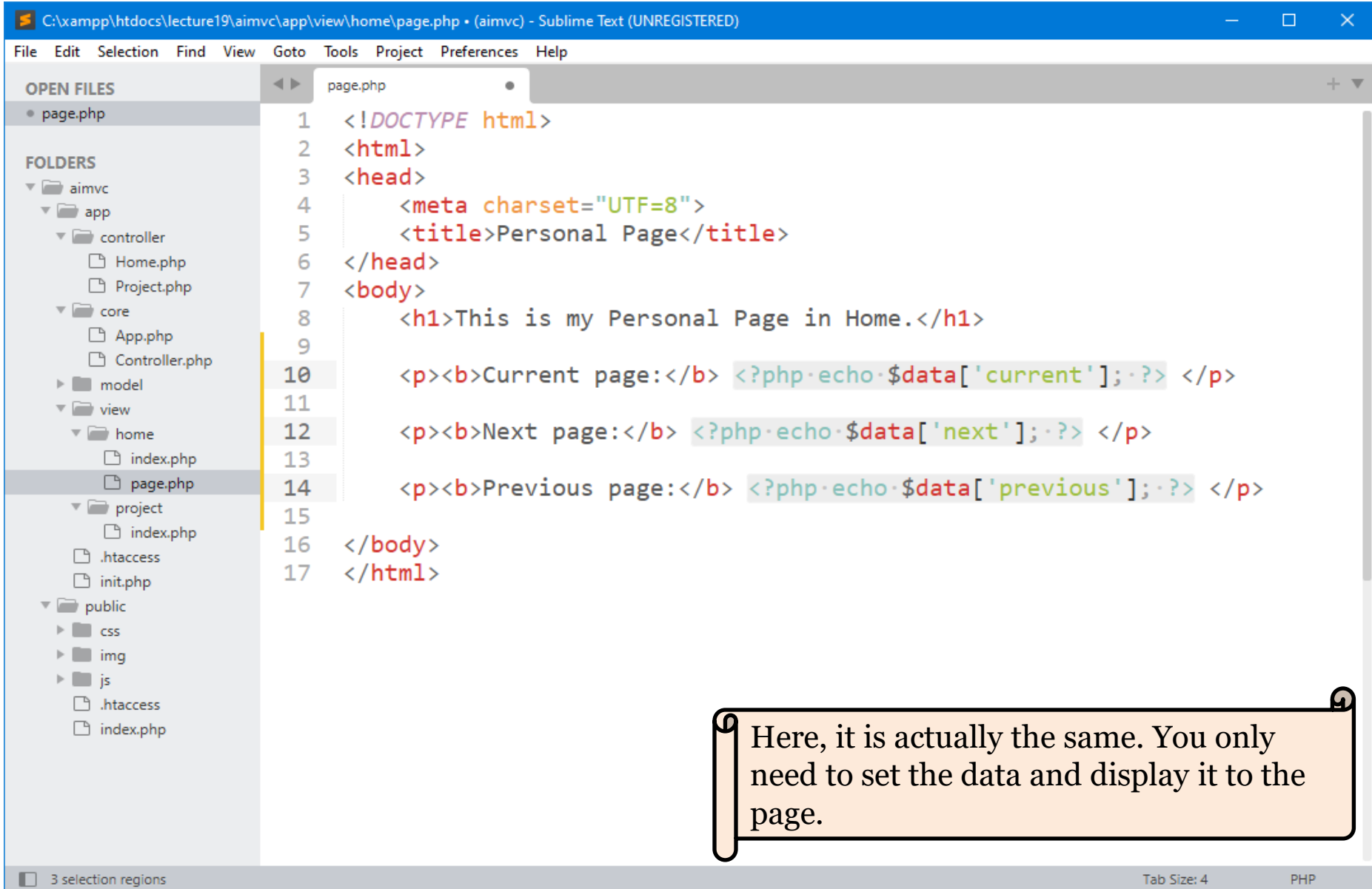
The code on lines 10 and 12 uses the `<?php echo $data['name']; ?>` and `<?php echo $data['age']; ?>` syntax to output the values of `$data['name']` and `$data['age']` respectively. These two lines of code are highlighted with a red rectangular box. The left sidebar of the editor shows the project structure, including folders like `aimvc`, `app`, `controller`, `core`, `model`, `view`, and `public`, with the `index.php` file selected in the `view/home` directory.

Now, go to **index.php** page in the home view directory and modify by following this code. You are trying to view the **name** and **age** in your home page.

Line 15, Column 8

Tab Size: 4 PHP

4 – Also Modify *page.php* file in Home Directory



The screenshot shows the Sublime Text editor interface. The title bar indicates the file path is `C:\xampp\htdocs\lecture19\aimvc\app\view\home\page.php` and the editor is (aimvc) - Sublime Text (UNREGISTERED). The menu bar includes File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, and Help. On the left, the 'OPEN FILES' pane shows `page.php` as the active file. The 'FOLDERS' pane shows a tree structure: `aimvc` (expanded) contains `app` (expanded), which contains `controller` (expanded) with `Home.php` and `Project.php`, `core` (expanded) with `App.php` and `Controller.php`, `model`, `view` (expanded), `home` (expanded) with `index.php` and `page.php` (selected), `project` with `index.php`, `.htaccess`, and `init.php`, and `public` (expanded) with `css`, `img`, `js`, `.htaccess`, and `index.php`. The main editor area displays the code for `page.php`, which is an HTML document with PHP code for displaying data. The code is as follows:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Personal Page</title>
6 </head>
7 <body>
8     <h1>This is my Personal Page in Home.</h1>
9
10    <p><b>Current page:</b> <?php·echo·$data['current'];·?> </p>
11
12    <p><b>Next page:</b> <?php·echo·$data['next'];·?> </p>
13
14    <p><b>Previous page:</b> <?php·echo·$data['previous'];·?> </p>
15
16 </body>
17 </html>
```

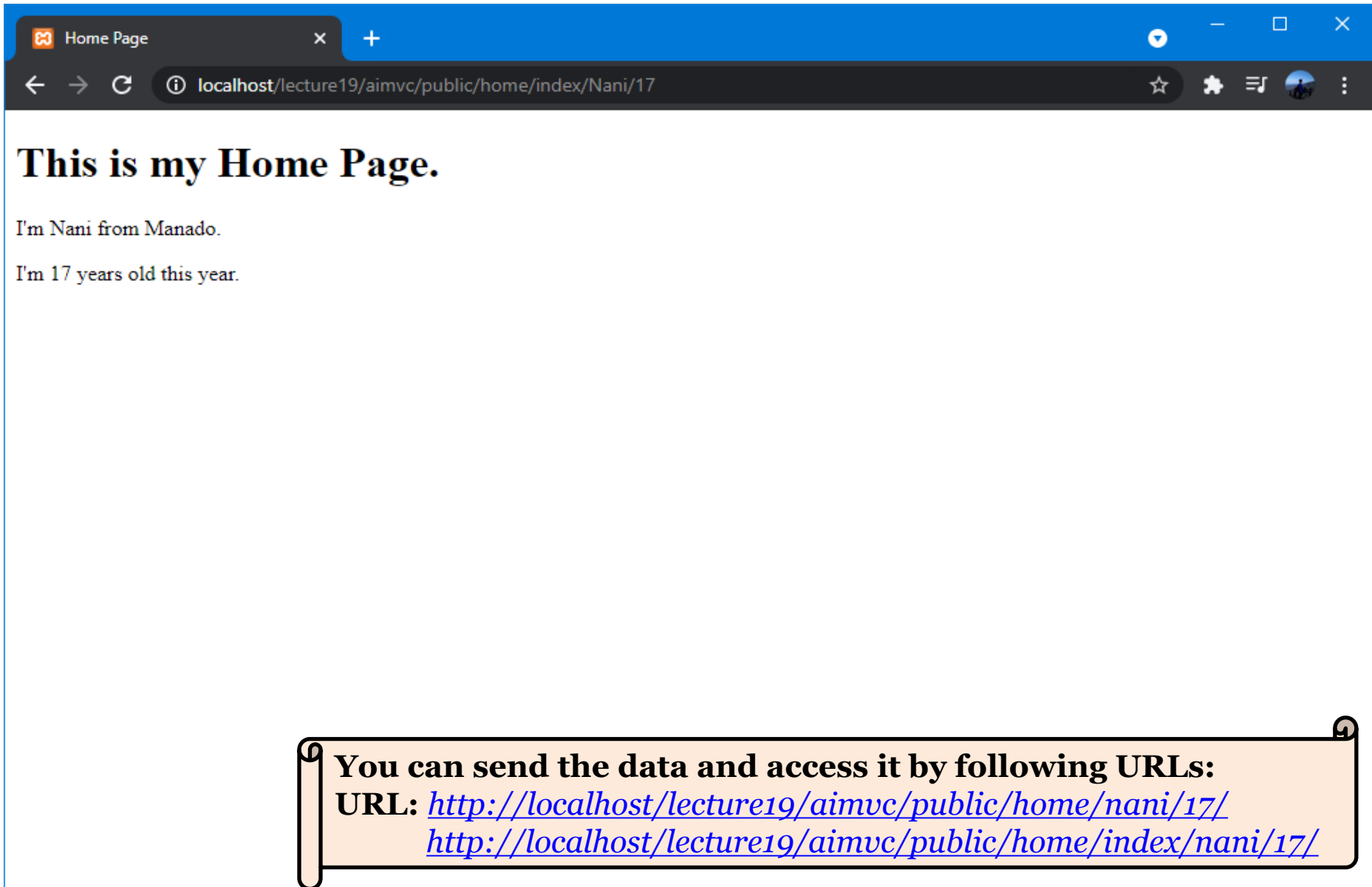
A callout box in the bottom right corner contains the text: "Here, it is actually the same. You only need to set the data and display it to the page."

3 selection regions

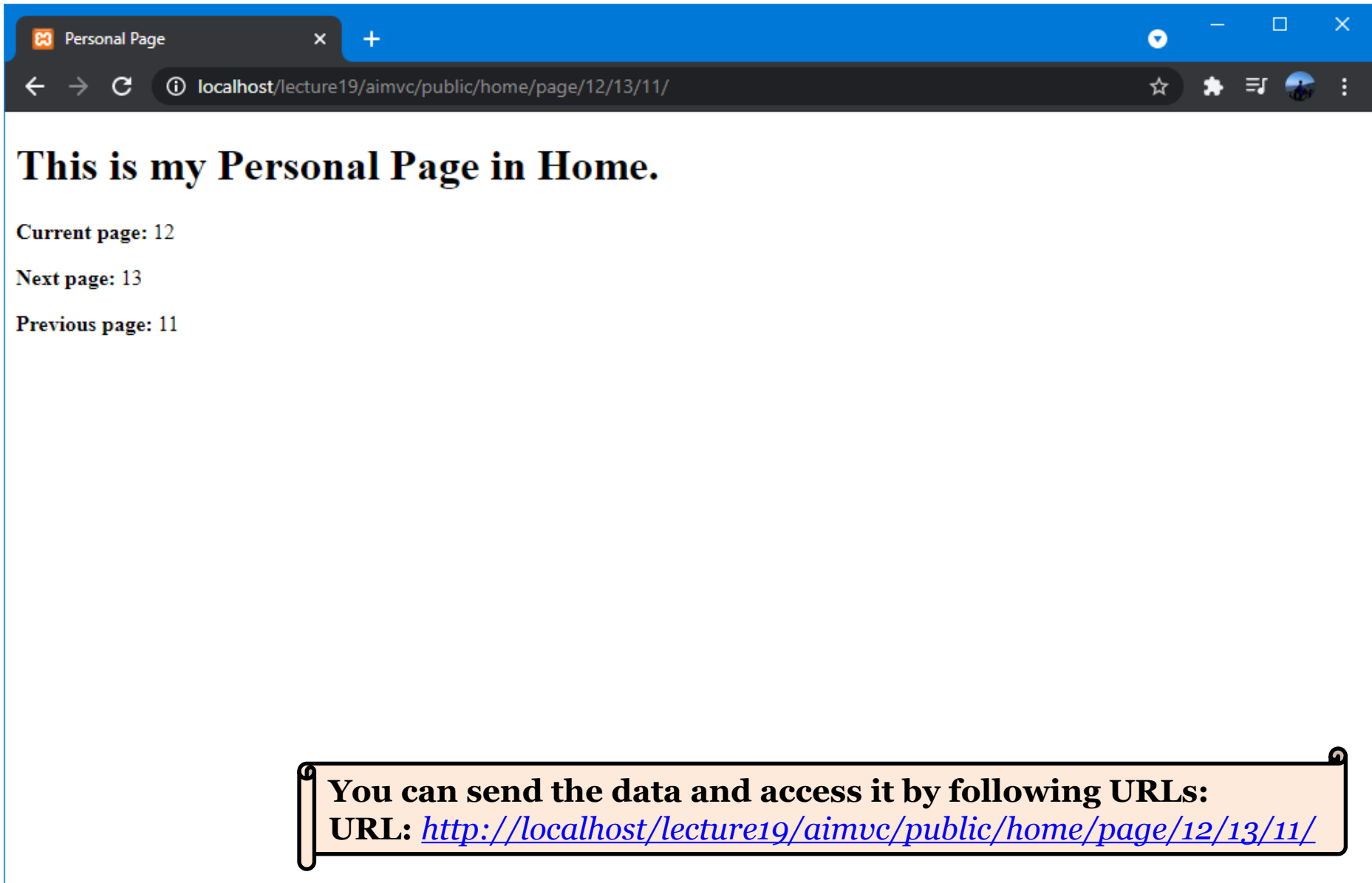
Tab Size: 4

PHP

5 - Testing ⇒ Home Index Page



6 - Testing $\Rightarrow$ Home Personal Page



The screenshot shows a web browser window with a single tab titled "Personal Page". The address bar displays the URL `localhost/lecture19/aimvc/public/home/page/12/13/11/`. The page content includes a heading "This is my Personal Page in Home." followed by three lines of text: "Current page: 12", "Next page: 13", and "Previous page: 11". At the bottom right, a callout box contains the text: "You can send the data and access it by following URLs: URL: <http://localhost/lecture19/aimvc/public/home/page/12/13/11/>".

This is my Personal Page in Home.

Current page: 12

Next page: 13

Previous page: 11

You can send the data and access it by following URLs:
URL: <http://localhost/lecture19/aimvc/public/home/page/12/13/11/>

A decorative border of green leaves and branches at the top of the page.

4 – Create Header & Footer Templates with *their* Custom Title

A decorative border of green leaves and branches at the bottom of the page.

HTML Header & Footer Templates

Header



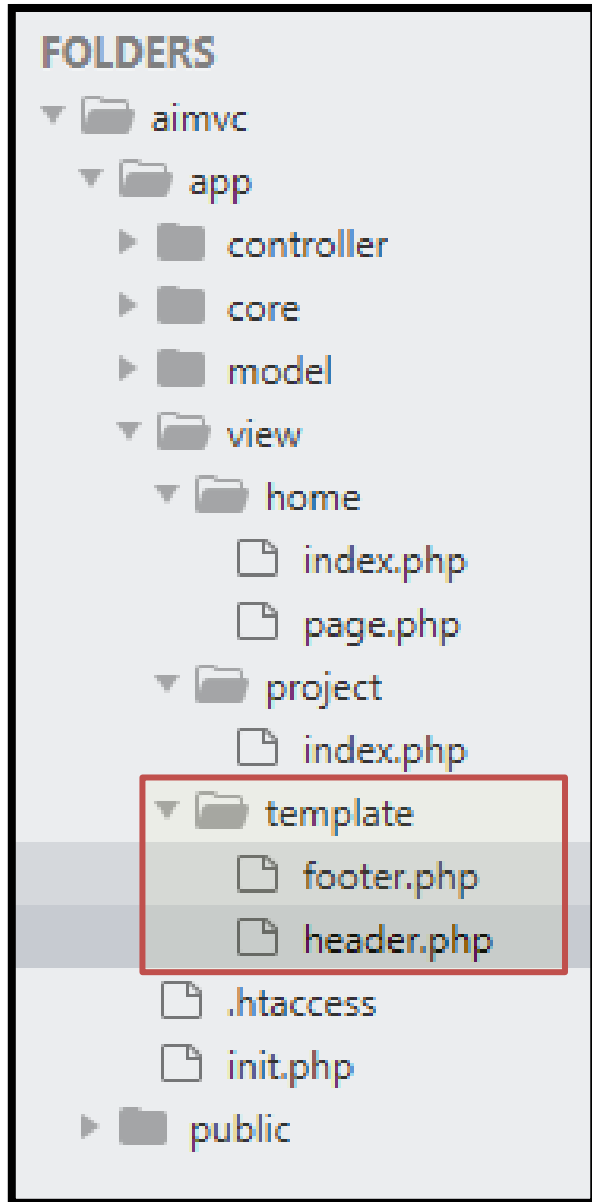
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Home Page</title>
6 </head>
7 <body>
8     <h1>This is my Home Page.</h1>
9
10    <p>I'm <?php echo $data['name']; ?> from Manado.</p>
11
12    <p>I'm <?php echo $data['age']; ?> years old this year.</p>
13
14 </body>
15 </html>
```

Footer



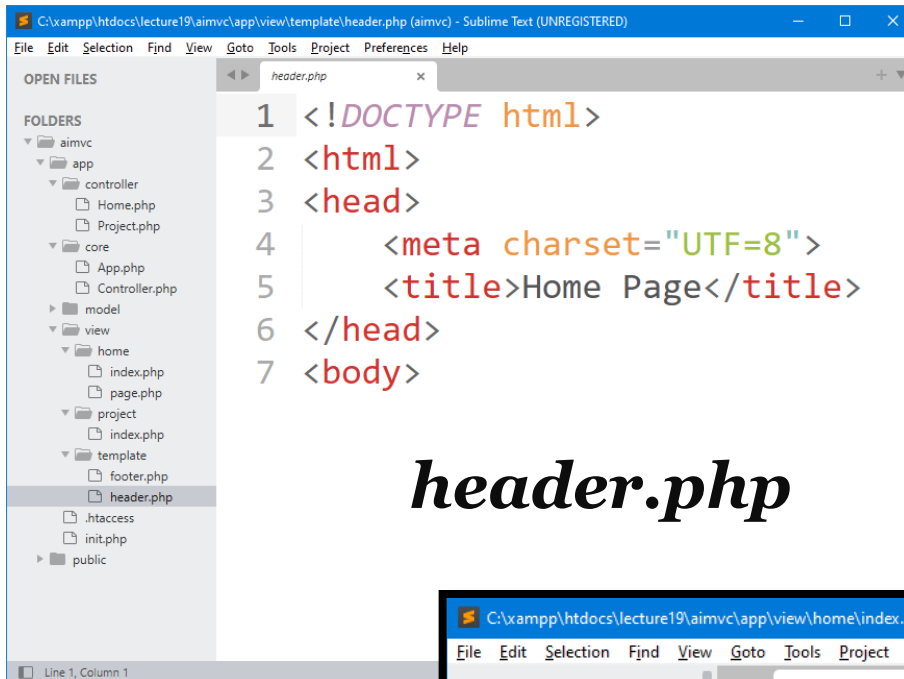
You can easily create sticky or fixed header and footer template using ***the CSS fixed positioning***. Simply apply the CSS position property with the value fixed in combination with the top and bottom property to place the element on the top or bottom of the viewport accordingly.

1 – Create Template Directory in View



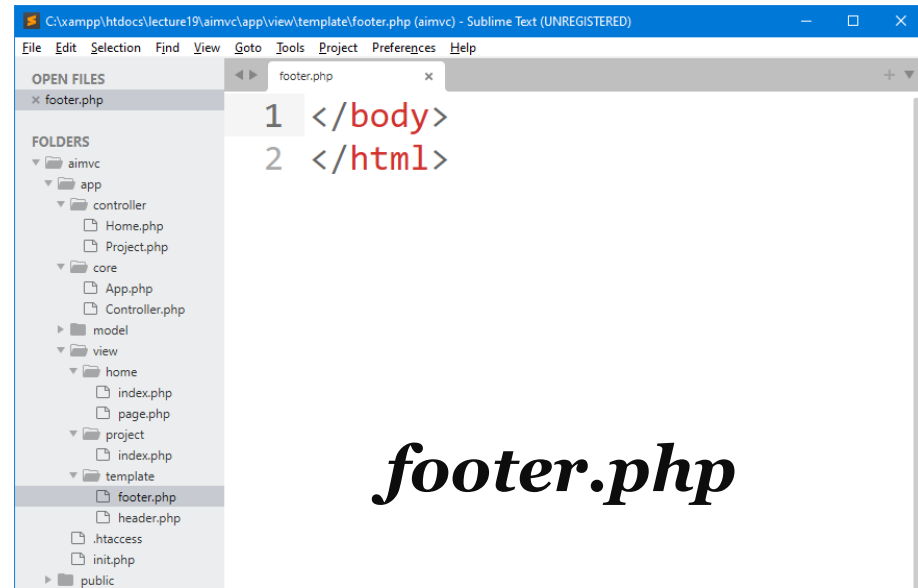
Create one subdirectory in view. This ***template directory*** can store ***header*** and ***footer*** templates and those files can be used as a template for all *php pages* in our framework project.

2 – Header, Footer & Content Sections



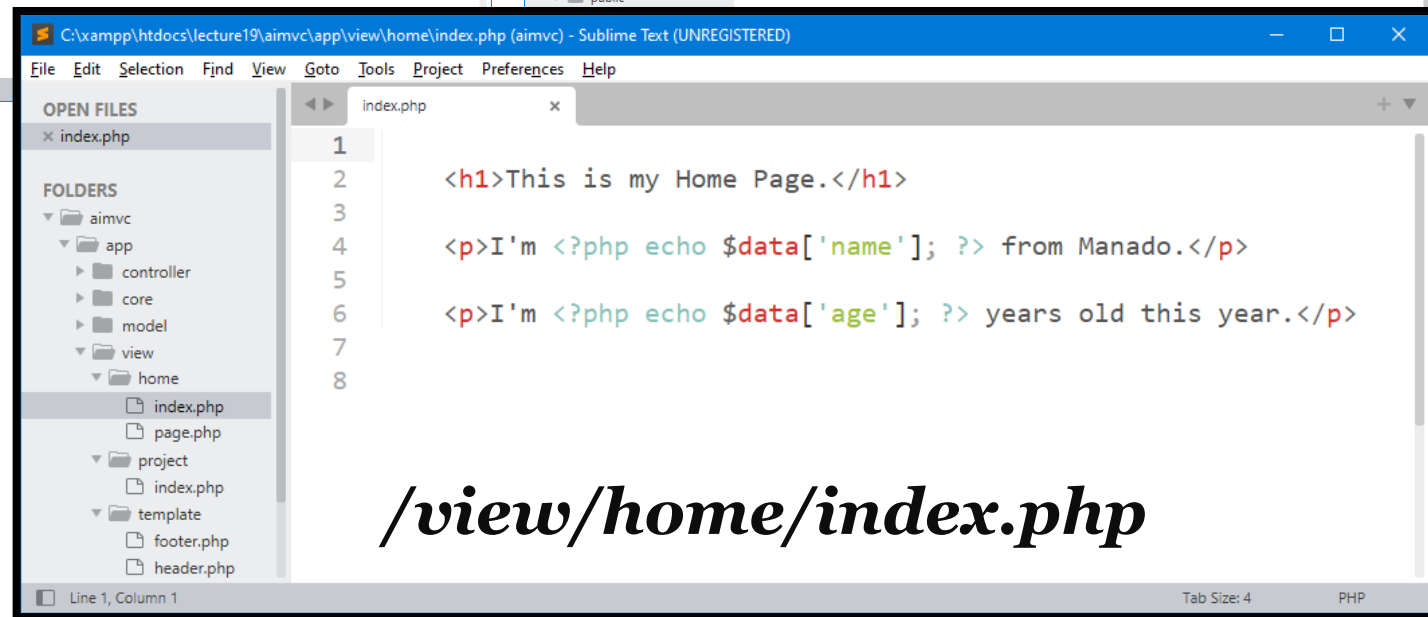
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Home Page</title>
6 </head>
7 <body>
```

header.php



```
1 </body>
2 </html>
```

footer.php



```
1
2 <h1>This is my Home Page.</h1>
3
4 <p>I'm <?php echo $data['name']; ?> from Manado.</p>
5
6 <p>I'm <?php echo $data['age']; ?> years old this year.</p>
7
8
```

/view/home/index.php

3 – Add Header & Footer in Controller

```
C:\xampp\htdocs\lecture19\aimvc\app\controller\Home.php (aimvc) - Sublime Text (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x Home.php

FOLDERS
aimvc
  app
    controller
      Home.php
      Project.php
    core
    model
    view
      home
        index.php
        page.php
      project
        index.php
      template
        footer.php
        header.php
      .htaccess
      init.php
    public

1  <?php
2  class Home extends Controller{
3      // Constructor
4      public function __construct(){
5
6      }
7
8      // Default method
9      public function index($name="Juan", $age="24"){
10         // Associative Arrays (arrays with keys)
11         $arr_data['name'] = $name;
12         $arr_data['age'] = $age;
13         // Display page and send data
14         $this->display('template/header');
15         $this->display("home/index", $arr_data);
16         $this->display('template/footer');
17     }
18
19     public function page($current_page=2, $next_page=3, $prev_page=1){
20         // Associative Arrays (arrays with keys)
21         $arr_data['current'] = $current_page;
22         $arr_data['next'] = $next_page;
23         $arr_data['previous'] = $prev_page;
24         // Display page and send data
25         $this->display('template/header');
26         $this->display("home/page", $arr_data);
27         $this->display('template/footer');
28     }
29
30     // Get status
31     private function _status(){
32         echo "This is home/status.";
33     }
34 }
35 ?>
```

Header

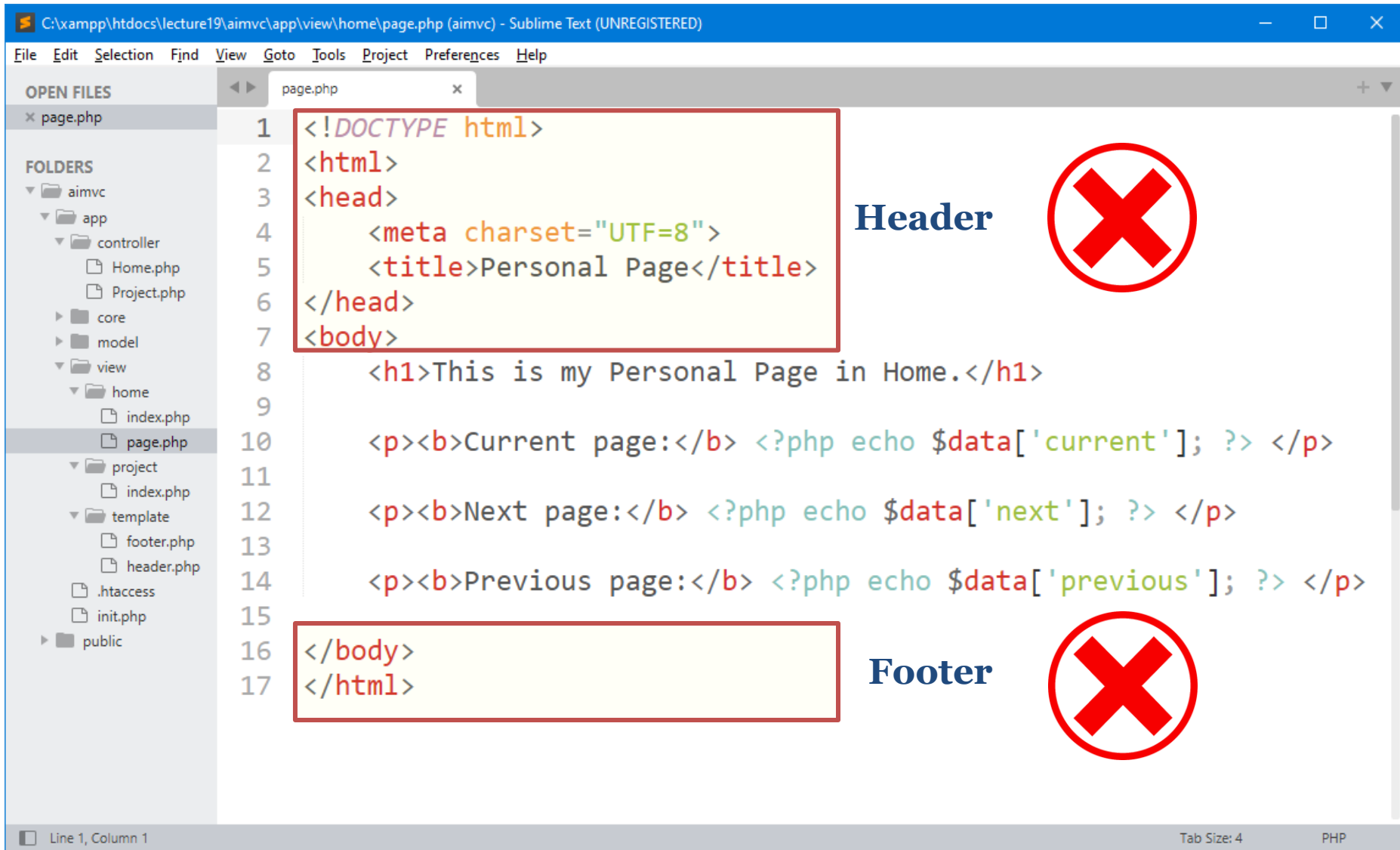
Footer

Header

Footer

4 – Don't Forget !!!

You may need to remove header and footer in **/view/home/page.php** file.



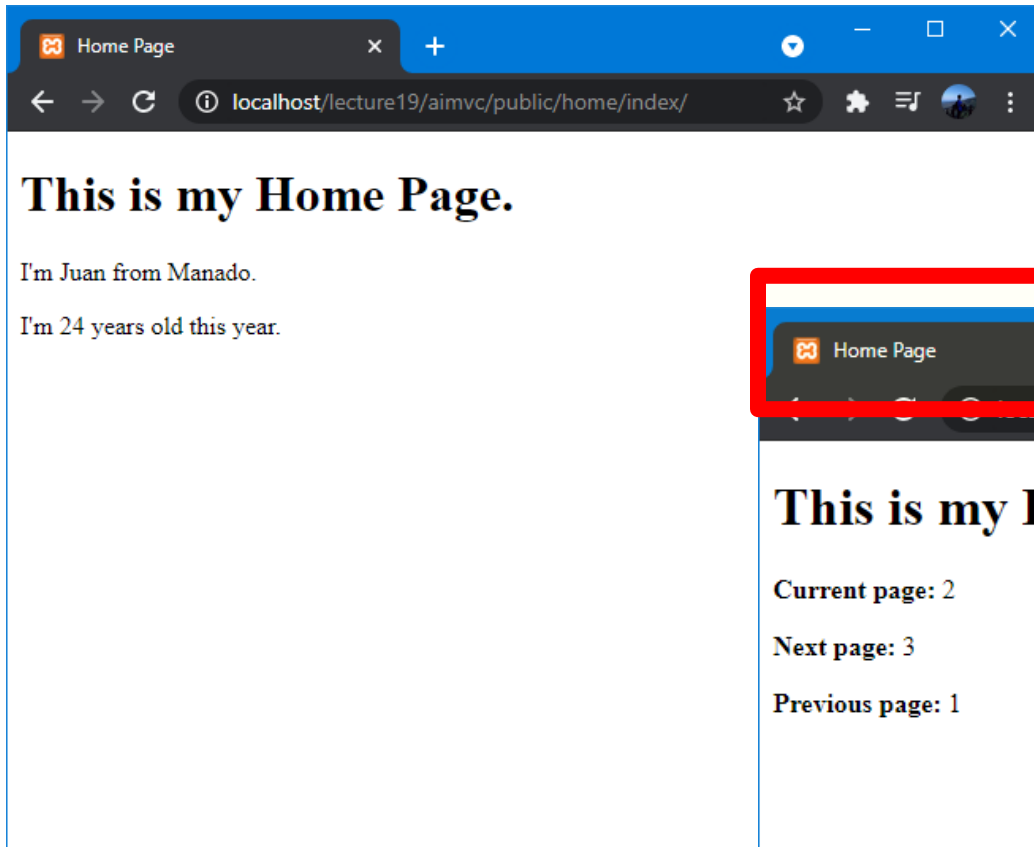
Header ❌

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title>Personal Page</title>
6 </head>
7 <body>
8     <h1>This is my Personal Page in Home.</h1>
9
10    <p><b>Current page:</b> <?php echo $data['current']; ?> </p>
11
12    <p><b>Next page:</b> <?php echo $data['next']; ?> </p>
13
14    <p><b>Previous page:</b> <?php echo $data['previous']; ?> </p>
15
16 </body>
17 </html>
```

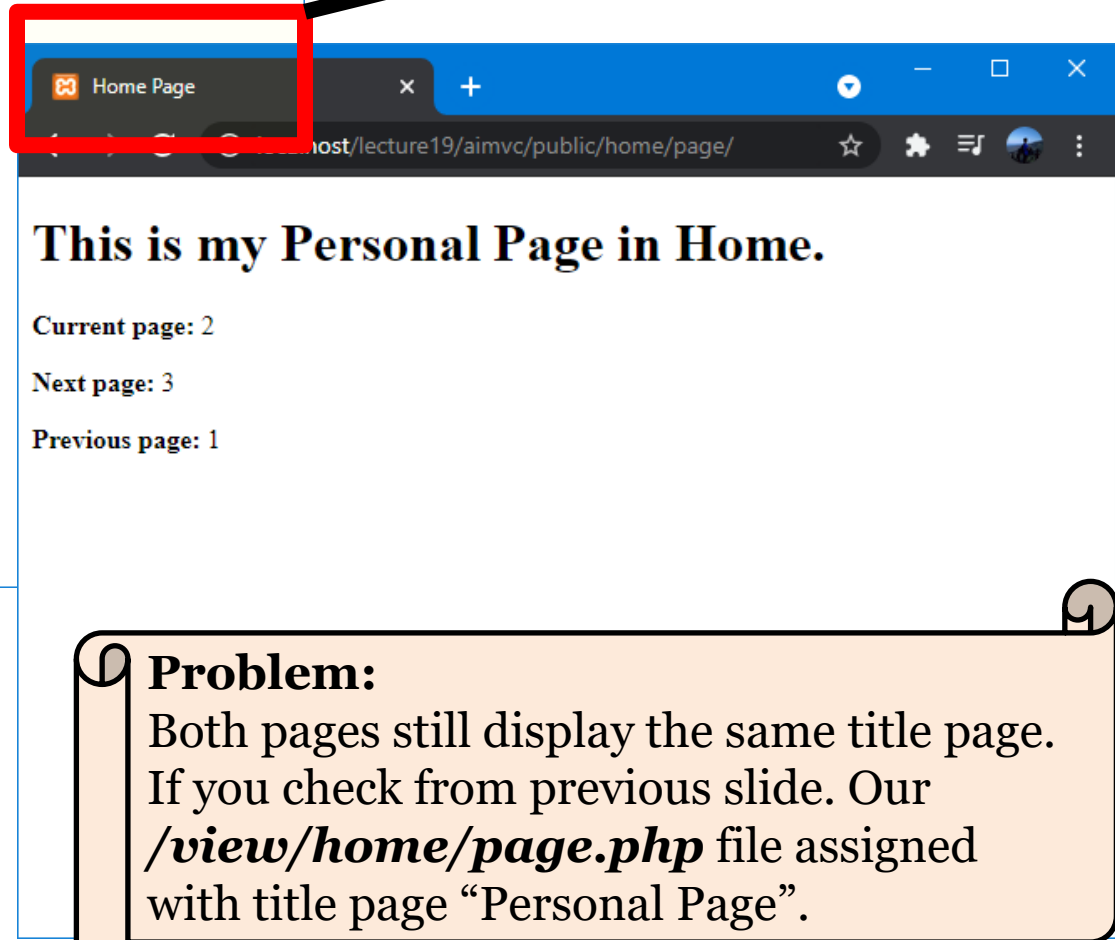
Footer ❌

Line 1, Column 1 Tab Size: 4 PHP

5 - Testing ⇒ Home and Personal Pages

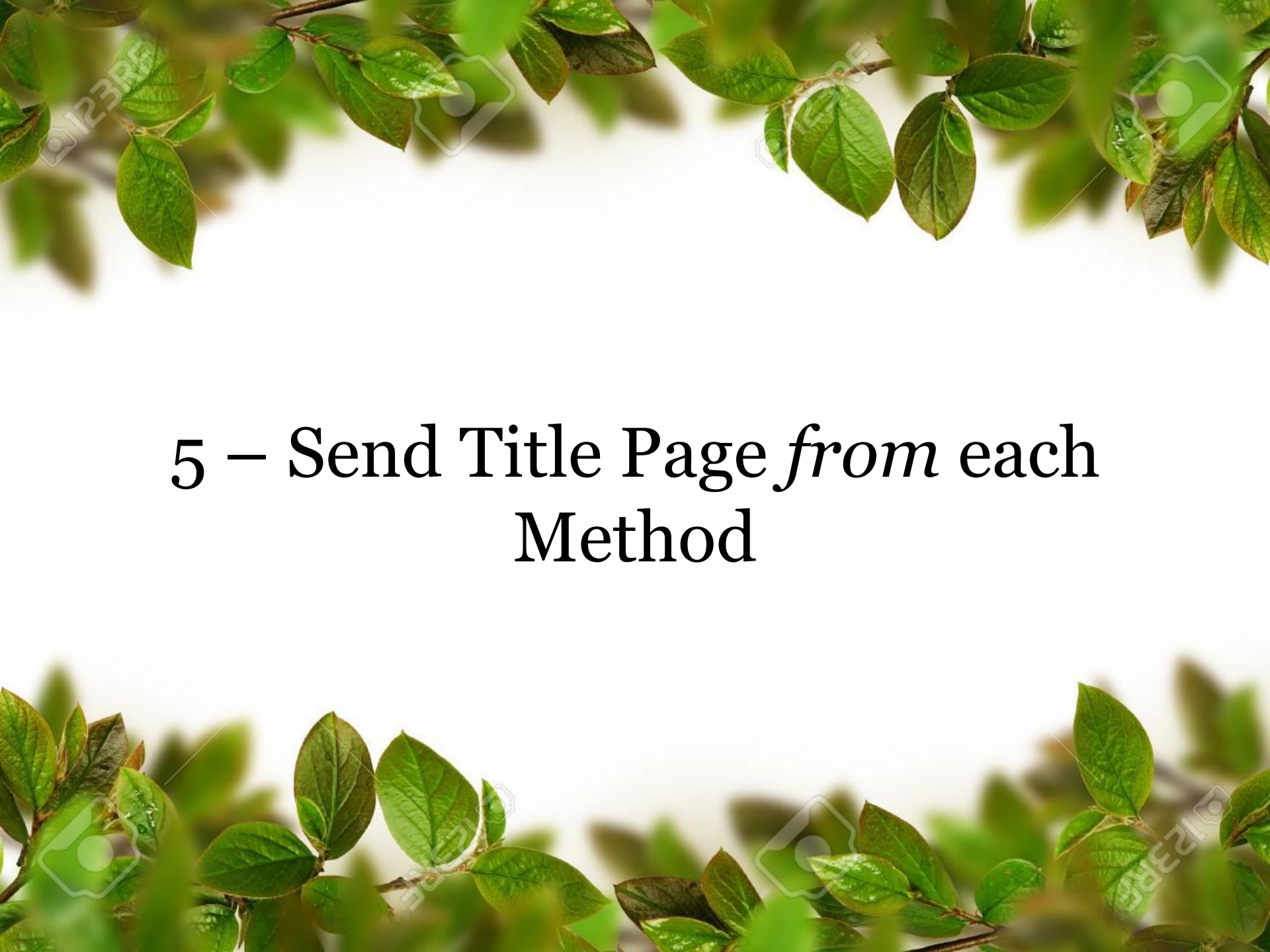


Wrong here !!!



Problem:

Both pages still display the same title page. If you check from previous slide. Our ***/view/home/page.php*** file assigned with title page "Personal Page".

The slide features a white background with green leaves framing the top and bottom edges. The leaves are vibrant green with some brown spots, suggesting they might be from a tree or shrub. The text is centered in the middle of the slide.

5 – Send Title Page *from* each
Method

1 – Send Title Page using Associative Arrays

```
C:\xampp\htdocs\lecture19\aimvc\app\controller\Home.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x Home.php

FOLDERS
aimvc
  app
    controller
      Home.php
      Project.php
    core
    model
    view
      home
        index.php
        page.php
      project
        index.php
      template
        footer.php
        header.php
    .htaccess
    init.php
    public

1  <?php
2  class Home extends Controller{
3      // Constructor
4      public function __construct(){
5
6      }
7
8      // Default method
9      public function index($name="Juan", $age="24"){
10         // Associative Arrays (arrays with keys)
11         $arr_data['name'] = $name;
12         $arr_data['age'] = $age;
13         $arr_data['title'] = "Home Page";
14         // Display page and send data
15         $this->display('template/header', $arr_data);
16         $this->display("home/index", $arr_data);
17         $this->display('template/footer');
18     }
19
20     public function page($current_page=2, $next_page=3, $prev_page=1){
21         // Associative Arrays (arrays with keys)
22         $arr_data['current'] = $current_page;
23         $arr_data['next'] = $next_page;
24         $arr_data['previous'] = $prev_page;
25         $arr_data['title'] = "Personal Page";
26         // Display page and send data
27         $this->display('template/header', $arr_data);
28         $this->display("home/page", $arr_data);
29         $this->display('template/footer');
30     }
31
32     // Get status
33     private function _status(){
34         echo "This is home/status.";
35     }
36 }
37 ?>
```

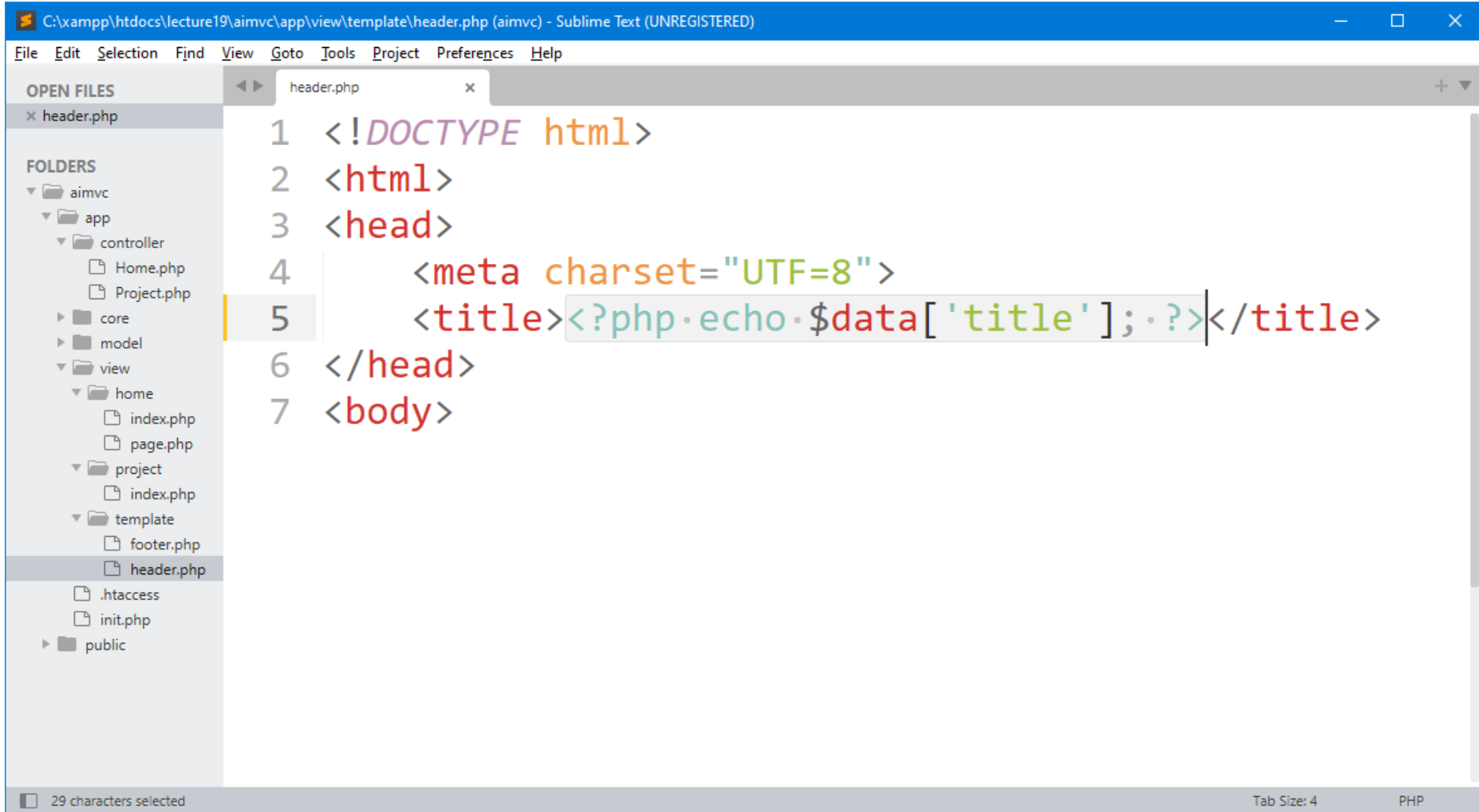
Title Page

Array data also need to send as a parameter in **display function**.

Title Page

Array data also need to send as a parameter in **display function**.

2 – Modify *header.php* file in Title Tag



```
C:\xampp\htdocs\lecture19\aimvc\app\view\template\header.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
header.php
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="UTF=8">
5     <title><?php·echo·$data['title'];·?></title>
6 </head>
7 <body>
```

29 characters selected

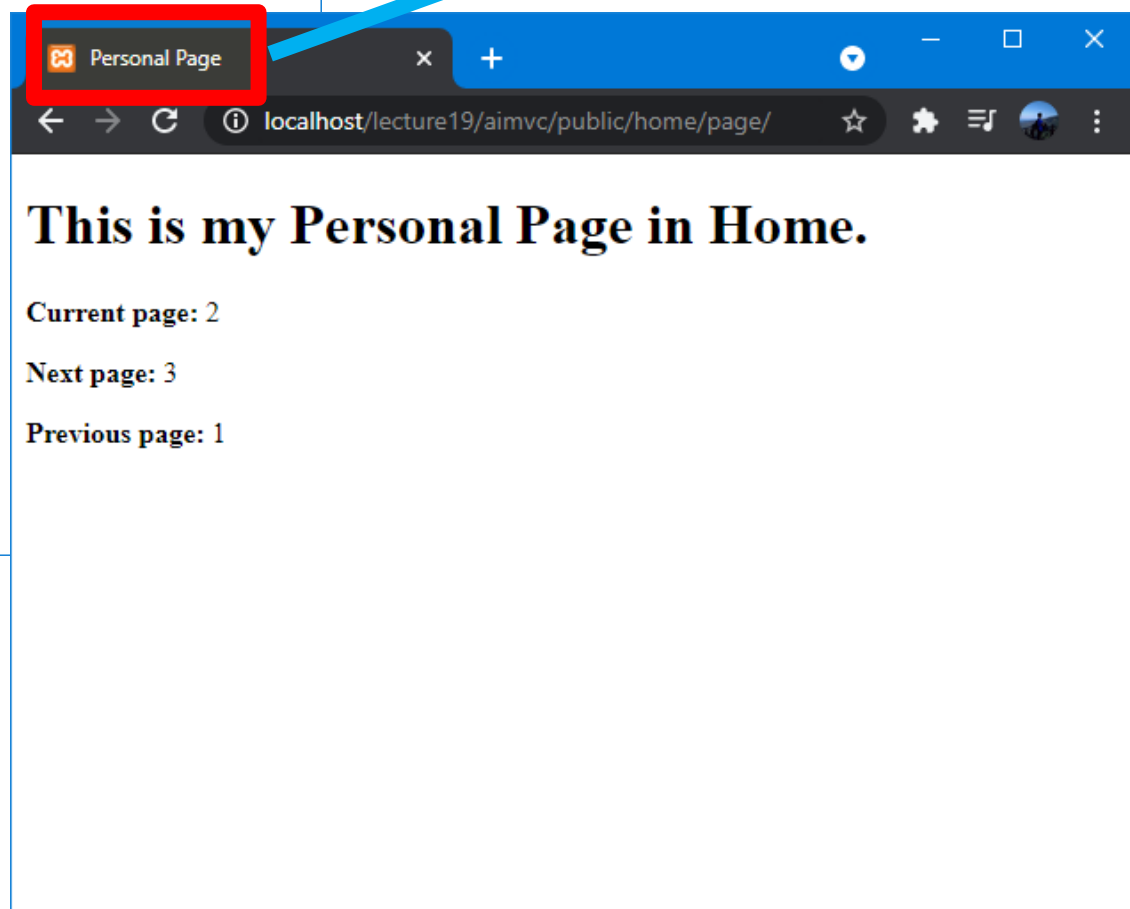
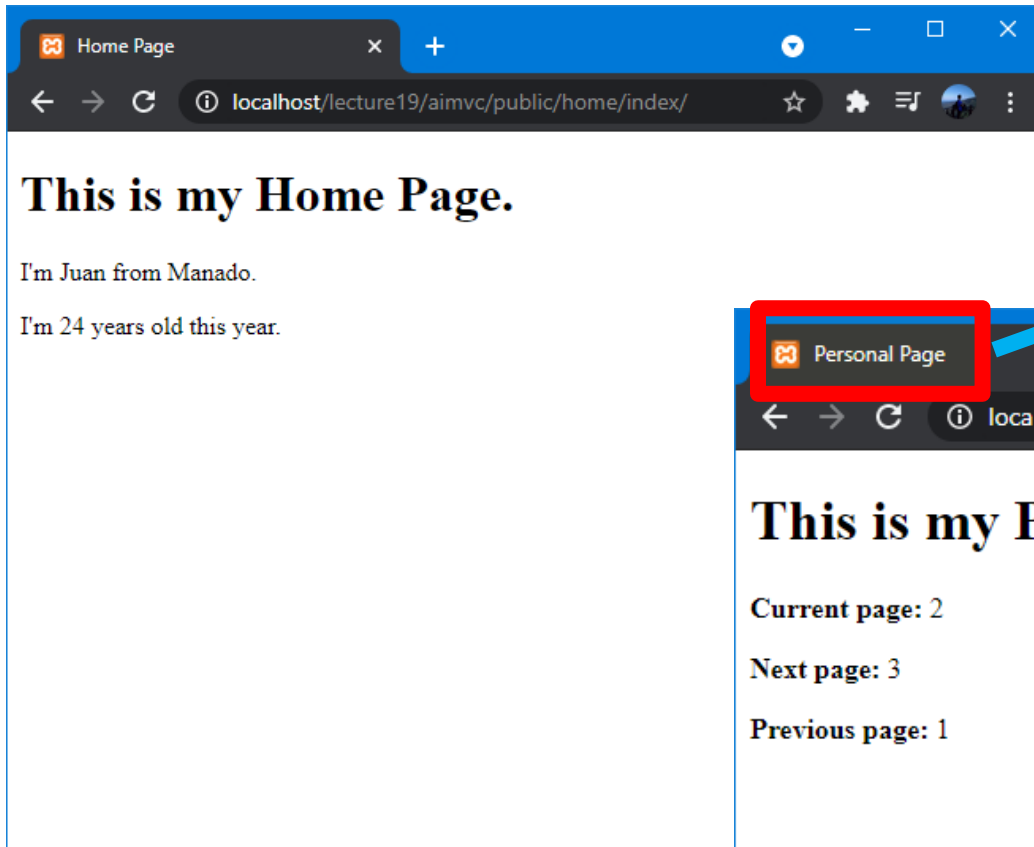
Tab Size: 4 PHP

Change following line of code:

From : <title>Home Page</title>

To : <title><?php echo \$data['title']; ?></title>

3 - Testing ⇒ Home & Personal Pages



It's correct Now !!!



Exercise *for* Students

**Pertama, lakukan uji
coba di local PC
masing-masing.**

Latihan untuk *Students*

Create directory : C:\xampp\htdocs\lecture22\
Create file : `_.php`

Buat semua contoh yang sudah di pelajari sebelumnya di *lecture* ini.

**Selanjutnya, upload
code Lecture #22 ke
sisi server untuk di
Demo.**

Create Directory in Server

[1] Masuk sebagai *administrator* atau *root*:

➤ ***sudo su***

[2] Buat project folder baru dengan perintah (Parent directory):

➤ ***sudo mkdir -p /var/www/html/lecture22***

[3] Konfigurasi permisi hak akses dan pemilik dari folder:

➤ ***sudo chmod -R 755 /var/www/html/lecture22***

➤ ***sudo chown -R semmy:semmy /var/www/html/lecture22***

[4] Restart Apache Web Server:

➤ ***sudo systemctl restart apache2***

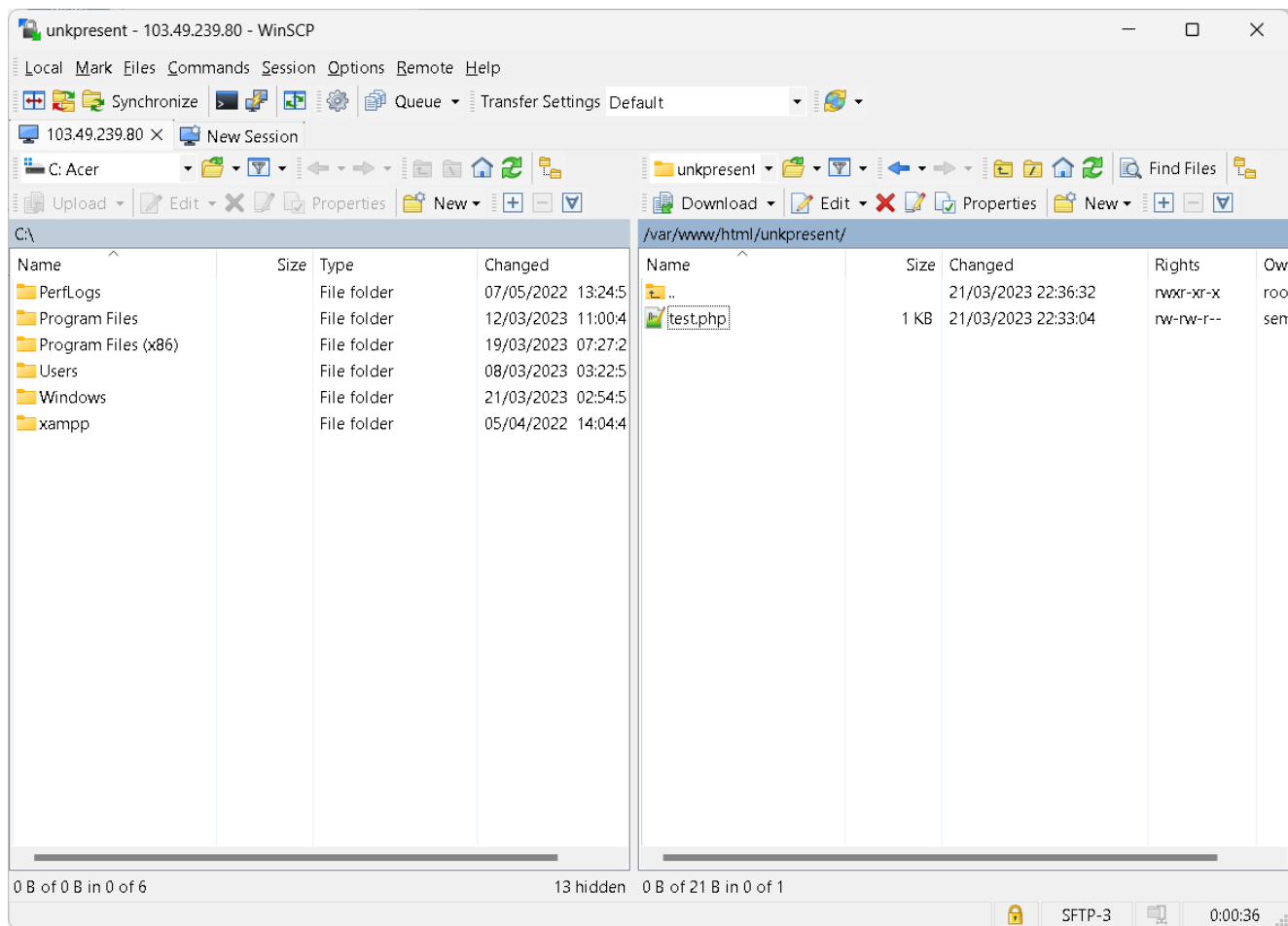
Note:

Ubah nama user “**semmy**” dengan nama *user* access ubuntu kalian.

```
root@unklab: /var/www/html
root@unklab:/home/semmy# sudo mkdir -p /var/www/html/unkpresent
root@unklab:/home/semmy# sudo chmod -R 755 /var/www/html/unkpresent
root@unklab:/home/semmy# cd ..
root@unklab:/home# cd ..
root@unklab:/# ls
bin  dev  home  lib32  libx32  media  opt  root  sbin  srv  tmp  var
boot  etc  lib  lib64  lost+found  mnt  proc  run  snap  sys  usr
root@unklab:/# cd var
root@unklab:/var# ls
backups  crash  local  log  opt  snap  tmp
cache  lib  lock  mail  run  spool  www
root@unklab:/var# cd www/
root@unklab:/var/www# ls
html
root@unklab:/var/www# cd html/
root@unklab:/var/www/html# ls
index.html  test.php  unkpresent
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 root root 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html# sudo chown -R semmy:semmy /var/www/html/unkpresent
root@unklab:/var/www/html# sudo systemctl restart apache2
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 semmy semmy 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html#
```

Upload Project via WinSCP

- 1) Buka WinSCP dan arahkan ke “/var/www/html/lecture22”.
- 2) Selanjutnya, upload **file code lecture#22** dari local PC dan ke directory server menggunakan WinSCP.
- 3) Buka via Browser.



Output ditampilkan menggunakan domain name, HTTPS dan bisa di akses secara public via Browser:

Contohnya:

<https://domainkelompok.site/lecture22/aimvc/public/>

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

